A Unified Explanation of Naturally Occurring Forces

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Abstract:

A modified quantized theory of relativity is used to explain the existence of the fundamental forces of nature in terms of the effects that divergent imaginary and real (IR) energy waves scattered off retarded (past) asymmetric distributions of elementary particles have on the symmetry of the retarded boundary conditions and the converging imaginary and real (IR) energy waves that give rise to elementary particles and that make each elementary particle and the observable universe in which they exist unique.

Résumé:

Une théorie quantique de la relativité, modifiée, est utilisée pour expliquer l'existence des forces fondamentales de la nature en termes d'effets que les ondes énergétiques imaginaires et réelles (IR) divergentes dispersées par des distributions asymétriques retardées (passées) de particules élémentaires exercent sur la symétrie des conditions aux limites retardées et sur les ondes énergétiques convergentes, imaginaires et réelles (IR), qui donnent naissance aux particules élémentaires et qui rendent uniques chaque particule élémentaire et l'univers observable dans lequel elles existent.

Introduction:

In a video of a Richard Feynman lecture in which he explained the existence of the repulsive electric force acting between two electrons in terms of the exchange of virtual photons, an attendee naively asked him to explain how two oppositely charged particles attracted each other if the electric force was due to the exchange of virtual photons. He responded to this naïve question by saying something along the lines of, 'the key is to not think of the exchange of virtual particles in terms of classical collision theory, i.e., as the collision of billiard balls, or the throwing and catching of baseballs, that it was far more complicated than that,' which seemed to me at that time to be an attempt to evade answering the question, because in his blackboard demonstration that is exactly the analogy that he used to describe the underlying cause of the repulsive force acting between two electrons.

At that time I thought that the answer he gave came up short because if the exchange of virtual photons could knock apart two particles with the same electric charge then it seemed to me that there would have to be a difference between the virtual photons exchanged between opposite electric charges and the virtual photons exchanged between like electric charges, which seemed ridiculous because as far as anyone has been able to tell 'photons are photons' and there is no evidence to support the hypothesis that there exists a field of negative energy virtual photons that would make it possible for two elementary particles exchanging such negative energy photons to move in a direction that is opposite the direction one would anticipate for particles exchanging, emitting, reflecting, scattering, positive energy quanta, i.e., that would increase the momentum of the two particles relative to each other in such a way that the two particles would appear to accelerate towards each other.

Particles Trapped in an Expanding Infinite Potential Well:

While taking an introductory course in quantum mechanics I was instructed as to how to use the de Broglie concept of matter waves and the time independent Schrödinger equation to calculate the most probable position and the energy of the stationary states of an elementary particle "trapped" (ignoring tunneling) inside a potential well. By the end of that course I was introduced to the time dependent solution to the problem of finding the most probable position of an elementary particle in a one-dimensional potential well when the left and right boundaries of the one-dimensional potential well appeared to be asymmetrically adiabatically moving relative to an elementary particle contained in it.[1] The solution to this problem, which is called the adiabatic approximation, revealed that the most probable position of an elementary particle contained inside an asymmetrically adiabatically expanding potential well (ignoring for now tunneling) appeared to adiabatically shift towards the boundary that appeared to be moving away from an elementary particle trapped within with the greatest velocity when the potential well was expanding and away from the boundary with the greatest velocity when the one-dimensional well was contracting —That in the special case when relative to an observer one boundary of the one-dimensional potential well appeared to be static while the other boundary appeared to be adiabatically moving towards or away from the boundary that appeared to be static; the most probable position of the elementary particle contained in the one-dimensional potential well would appear to adiabatically move away from what appeared to be the static boundary of the one dimensional potential well and towards what appeared to be the adiabatically receding boundary of the potential well or to move towards what appeared to be the static boundary of the one-dimensional potential well and away from what appeared to be the adiabatically approaching boundary of the potential well —And when the adiabatic change in the size of the potential well is restricted i.e., the potential well adiabatically transitions from some

initial size a_i to some final size a_f, the Hamiltonian of an elementary particle in the *n*th eigenstate of H^i in the original potential well will, under the Schrödinger equation, adiabatically transition to the *n*th eigenstate of H^f in the larger or smaller potential well.

In the traditional one-dimensional depiction of the potential well, the complex matter waves that give rise to elementary particles are naively depicted as moving from the left and right boundaries of the potential well and then back to the same boundaries that separate the inside of the potential well from the outside[2] (See figure 1).

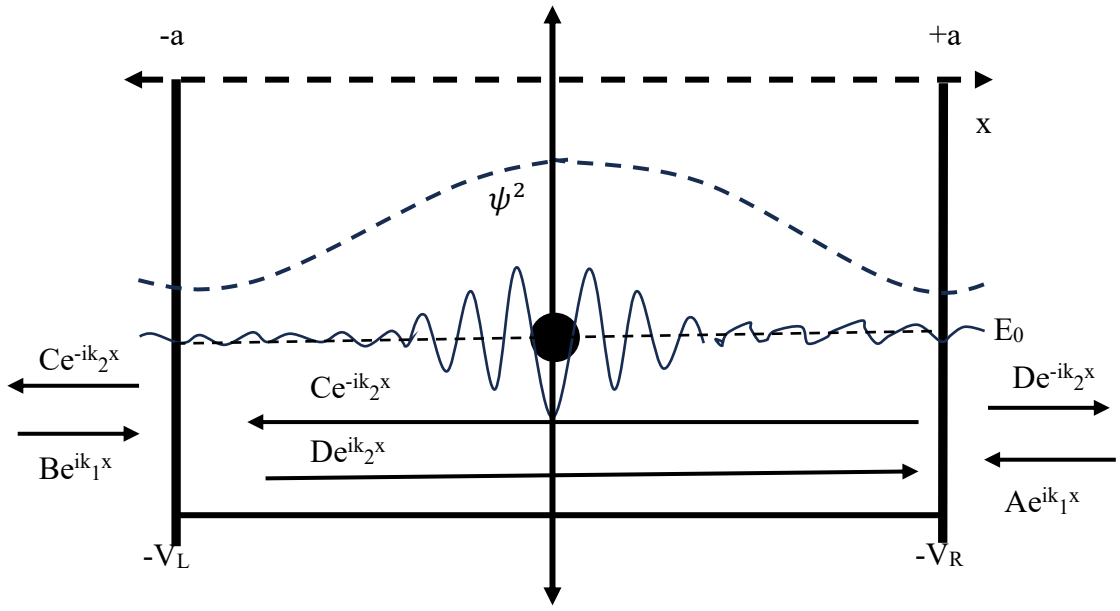


Figure 1

Traditional representation of a particle i.e., matter waves in a one-dimensional square potential well.

This naïve depiction has several serious problems associated with it. Firstly, it does not convey the fact that in an observable universe in which the speed of light restricts the speed at which information can be transmitted from one place to another the boundaries of the potential well will appear to exist in the past relative to all observers in the observable universe. Secondly, the velocity with which the boundaries of a potential well appear to be moving towards or away from an elementary particle contained in an adiabatically contracting or expanding potential well is dependent upon the relative velocity of an observer, except in the special case when the boundaries appear to be moving at the speed of light or faster. Thirdly, in an observable universe in which particles and waves are required to temporally move from the past to the future relative to all observers in it, the matter waves that give rise to elementary

particles and determine their most probable position and group velocity cannot simply be depicted as returning from whence they came thus making it impossible to trap a particle between two ever present potential boundaries in such a way that matter waves that arrive from the past boundary of the observable universe can return to the same past boundary from whence they came. Fourthly, relative to a particular elementary particle or ensemble of elementary particles, the presence of other retarded elementary particles or groups of elementary particles in a potential well will affect the symmetry of the potential well (make one side appear to be farther away) and also affect the phase velocity of the matter waves that give rise to the elementary particles contained in the potential well, which in turn will alter the most probable relative positions and group velocities of the elementary particles contained in the potential well.

In both the theory of quantum electrodynamics and relativistic quantum field theory, elementary particles are considered as either being bound or free. Although the idea of a free elementary particle greatly simplifies the mathematics of both theories, in fact all elementary particles exist in a potential well called the universe, which implies that asymmetries in the distribution of potential energy about an elementary particle when taken in conjunction with the relativistic time delay of signals as they pass through and around asymmetrically arranged concentrations of imaginary and real forms of potential energy (IR potential energy) should give rise to asymmetric IR boundary conditions which quantum theory predicts should give rise to a change in its group velocity, i.e., depending on the group velocity of an elementary particle, acceleration or deceleration towards what appears to be the longest, oldest, coldest, least dense, most red shifted, most entropic region of spacetime and away from what appears to be the shortest, youngest, hottest most dense, least red shifted, least entropic region of spacetime.

In classical theory, the rest electric charge and rest mass of elementary particles are taken to be fixed quantities but in quantum field theory the intrinsic naked "rest" electric charge and mass of an elementary particle are taken to be infinite, and the observed finite magnitude of an elementary particle's electric charge is attributed to the shielding of that particle's infinite electric charge by the polarization of a fluctuating field of virtual particles —A ubiquitous fluctuating field comprised of virtual photons that are being transformed into virtual particle anti-particle pairs and then back into virtual photons in an unmeasurably small amount of time; while the rest mass/energy of an elementary particle is defined by the strength with which it couples to a virtual field of Higgs bosons and the Heisenberg uncertainty relationships.[3]

In the present day macroscopic observable universe, the electric charge and rest mass of elementary particles appear to be constants of nature and in the classical limit the magnitude of the electric and gravitational interactions are observed to be inversely proportional to the square of the distance separating the charges/masses. Presently there is no evidence to suggest that the magnitude of the rest electric charge and rest mass of elementary particles are functionally related to the age/size of the universe, but if the density of an expanding or contracting universe is directly proportional or inversely proportional to the age (size) of the universe then it seems reasonable to hypothesize that the density of the virtual fields that permeate spacetime are also directly proportional or inversely proportional to the age of the universe and because of this the mathematical relationships that are currently being used to describe the behavior of elementary particles in the observable universe may need to be augmented to account for temporal variations in the density of the fields that determine the magnitude of the electric charge and rest mass of elementary particles.

The idea that the magnitude of the electric charge of elementary particles is a time dependent functions of the shielding density of the retarded virtual field of particles that are hypothesized to surround them, implies that in an inflating observable universe the electric charge of elementary particles began to increase after the emergence of the observable universe and continued to increase to its present day value or in the case of a deflating observable universe, the electric charge of elementary particles began to decrease from an infinite value and fell to its present day value as spacetime contracted. Note that although the present-day scientific consensus is that the universe is inflating, for what I think are very sound reasons as outlined in a paper titled "**Does the cosmological red shift have a gravitational component**"[4] I am very skeptical of the conclusion that the universe is inflating at the rate that it is presently thought to be inflating and based upon the hypothesis examined in the above mentioned paper, for all anyone knows, it could actually be contracting and only appear to be inflating. In either case, the universe could be bouncing between these two scenarios and it is for this reason that I think the possibility of a contracting observable universe needs to be mentioned here.

Additionally, if it is assumed that elementary particles are manifestations of interacting matter and charge waves that originate on the boundary of an expanding/inflating, or contracting/deflating observable universe, and if these suspected energy density variation also give rise to variations in the virtual fields that permeate spacetime, then there exits the possibility that the relative position and momentum of elementary particles could be affected by the time it takes signals to traverse regions of retarded spacetime containing different imaginary and real (IR) potential energy densities —IR energy densities that could induce relativistic time delays and phase shifts that could affect the IR energy waves (what used to be called matter waves) that

give rise to what appears to be a pixilated observable universe in which the behavior of the elementary particles in it appears to be influenced by magical or mysterious forces.

Consideration also needs to be given to the quantum understanding that elementary particles trapped in a potential well have a probability of tunneling across the boundaries of a potential well, which in this case is the observable universe implies that the rate at which elementary particles appear to tunnel in and out of the observable universe should be related to the energy of the elementary particles in it and the size of the potential well that contain them i.e., the number of elementary particles that tunnel into and out of the potential well we call the observable universe should be infinitesimally small when the observable universe is very large and cold and very large when it is very small and hot. This gives rise to the possibility of an observable universe in which elementary particles are virtually tunneling in and out of existence at rates that are dependent upon the state it is in at a particular time, which implies that charge parity, and baryon conservation might not be constant with respect to cosmological time and may not exist at all in imaginary or virtual regions of spacetime, which could give rise to what appears to be mysterious repulsive and attractive forces that could affect the rate at which the observable universe appears to be expanding or contracting relative to observers trapped in it.

If it is assumed that asymmetries in the IR potential energy density of the retarded spacetime regions that appear to surround a particular elementary particle will make the observable universe appear to be asymmetrically inflating/deflating relative to observers comoving with that elementary particle, i.e., make the universe appear to be older, colder, bigger, less dense, more red shifted, more entropic in the retarded direction of the largest concentration of IR potential energy and younger, hotter, smaller, more dense less red shifted, less entropic in the retarded direction with the least concentration of IR potential energy, then the possibility

exists that asymmetries in the density of the IR potential energy in the retarded volume of spacetime that appears to surround an elementary particle or an ensemble of elementary particles could make it seem that mysterious or magical forces are somehow causing that elementary particle or ensemble of elementary particles to accelerate/decelerate towards what is measured to be the oldest, coldest, biggest, least dense, most red shifted, most entropic regions of the universe, and away from what is measured to be the youngest, hottest, smallest, most dense, least red shifted, least entropic regions of the universe.

Although the observable universe appears to be relatively isotropic on the cosmic scale obvious asymmetries exist in the distribution of IR potential energy in the retarded regions of spacetime that appear to surround elementary particles and although quantum theory predicts that these local asymmetries will give rise to a ubiquitous quantum attractive force in an inflating universe, preliminary naive calculations of the magnitude of this attractive force for elementary particles contained in an asymmetrically but nearly symmetric adiabatically expanding potential well the size of the observable universe falls short of the magnitude of the gravitational, nuclear, electroweak forces that are known to be responsible for driving the evolution of the universe. Thus, much more will need to be taken into consideration in order explain the elementary forces of nature in terms of the effects that asymmetric distributions of retarded IR potential energy have on the most probable position $\langle x \rangle$, momentum $\langle p \rangle$, and kinetic energy $\langle K.E. \rangle$, of particles in the observable universe.

Types of Boundaries.

In order to model the effects that asymmetric retarded distributions of IR potential energy have on the most probable position and group velocity of elementary particles contained within the boundaries of an adiabatically expanding/contracting universe, it will be necessary to identify

the boundaries that separate the universe into sub-luminal, luminal and virtual regions of spacetime. The following list contains six boundaries that will need to be considered when constructing models of spacetime.

1) The luminal boundary, which restricts how fast information can be transmitted from one point of the observable universe to another and that separates observable regions of the universe from the unobservable virtual regions of the spacetime.
2) The Temporal boundary which requires that signals travel between two points in the observable macroscopic universe from the past to the present, i.e., it is not possible for macroscopic objects to travel backwards in time or to send information to past states of the macroscopic universe.
3) The Heisenberg Uncertainty principle which defines the limits of what can be known about the observable universe, and the intervals of spacetime where the distance separating two events appears to be null. This boundary is extended upwards by the sensitivity of the instruments that scientists use to observe the universe, and for this reason the universe is divided into three distinct regions a) the observable universe, b) the instrument restricted unobservable universe and c) the virtual unobservable regions of the universe.
4) The singular origin of spacetime.
5) The Schwarzschild boundary of a black hole which conceals the singularity containing interior region of a black hole from exterior observers.
6) The cosmic microwave background (CMB).

Quantum And Relativistic Considerations:

In the Special Theory of Relativity an elementary particle's relationship to events in the spacetime continuum is often naively represented by an imaginary-real (IR) three-dimensional slice of two four-dimensional light cones that share a common vertex, with the bottom half of the IR light cone used to represent a particle's connection to the influences of retarded manifestations of IR potential energy and the top half used to represents a particle's potential to act as a source of IR potential energy.[5] Since there is no credible scientific evidence to support the hypothesis that signals can travel from the future or the present to the past, i.e., that sequences of events can be altered after they happen and until such time that someone reveals evidence that Tachyon signals can be sent from the future or the present and received in the past, it will be assumed that the points P_1 and P_2 in figure 2 represent the nexus of signals that arrive at P_1 and P_2 from the past and depart towards the future state of the universe (see figure 2).

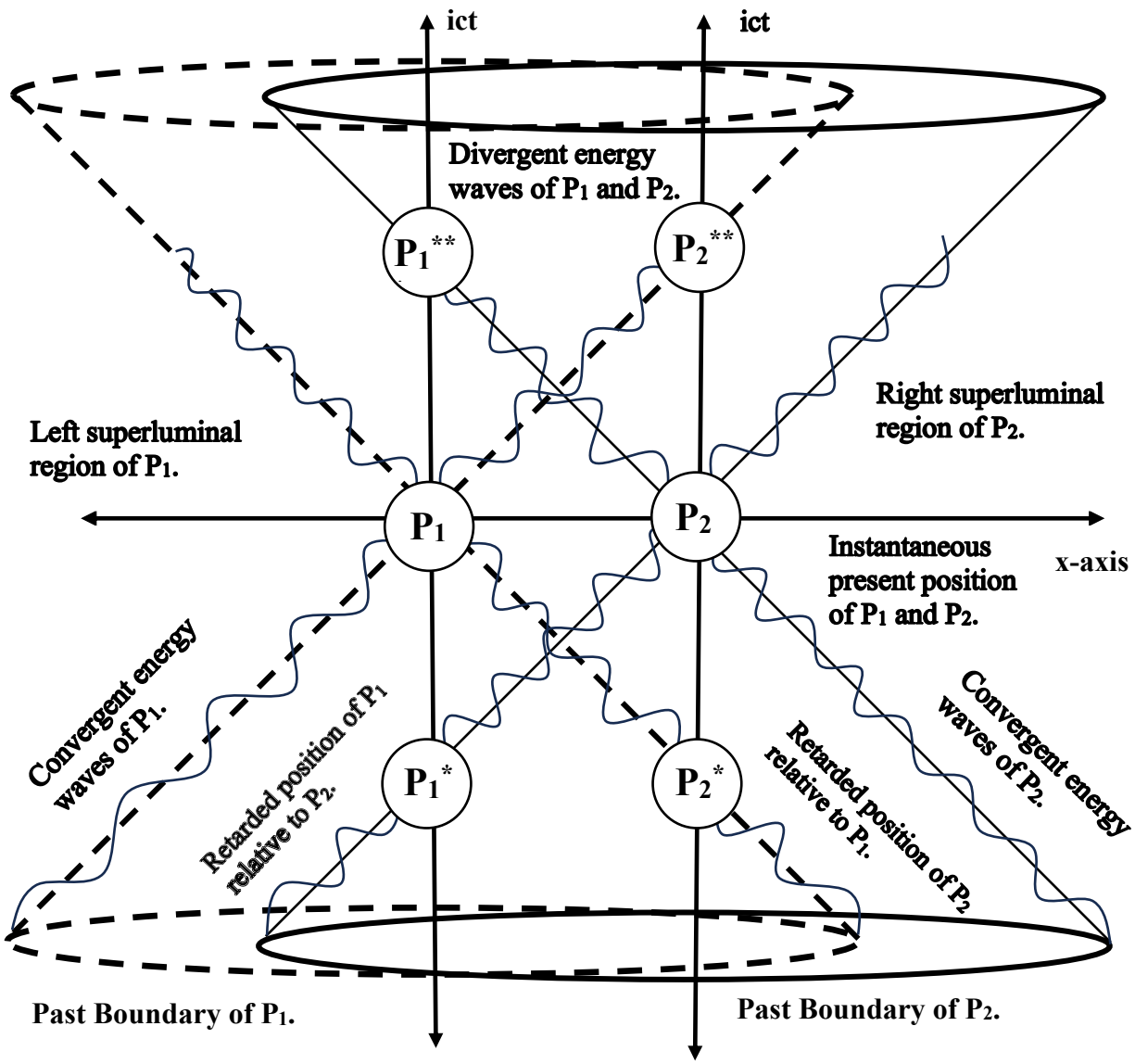


Figure 2

Traditional Fictitious Frame of reference representing the spacetime position of two comoving particles P_1 and P_2.

If the surface of the light cones in figure 2 represent the luminal boundaries of spacetime, then elementary particles, as represented by P_1 and P_2 in figure 2, can be thought of as unique manifestation of their quantized converging IR energy/information (particles/waves) that travel

from their unique singular boundary of the universe and are perturbed by asymmetries in the density of the IR potential energy they encounter as they travel from their unique boundary to their present position in the observable universe.

In modern relativistic theory, events in the observable universe are generally mapped with respect to four interdependent coordinate axes, three of which are usually defined to be orthogonal real coordinate axes and a fourth coordinate axis that is used to represent the dimension of time along which temporal coordinates are usually represented spatially by imaginary numbers in terms of *ict* dimensional units. Henceforth, mathematical systems that assign elementary particles and mathematical points in spacetime both real and imaginary coordinates will be called IR coordinate systems.

In the Special Theory of Relativity the IR metric that is used to calculate the distance between events in spacetime is defined to be pseudo-Euclidian, with spacetime distances expressed in terms of a pseudo-Pythagorean non Argandian IR distance function in which the square of imaginary temporal intervals give rise to negative real distances that when added to the square of real spatial intervals give rise to contracted and dilated distances and in the case of quantized electromagnetic waves, to null geodesics.[5]

Further examination of the relativistic light cone in figure 2 reveals the possibility that imaginary, super luminal, or virtual regions exist outside the luminal boundary of the observable universe, i.e., regions of the universe in which elementary particles can interact and exists in intervals of less than no observable time relative to observers in the observable universe. These unobservable regions of spacetime will be called null inner spaces.

Examining the light cone model that is used to depict these null inner super luminal regions of the universe reveals that events that occur in these superluminal regions are virtual

relative to an observer in outer regions, i.e., occur in virtual time intervals relative to observers outside the event horizon that separates the null inner regions or imaginary space like regions of the universe from the outer regions or imaginary time like regions of spacetime. Note, the virtual events that occur in the inner regions could occur in imaginary intervals of less than no observable time or in negative intervals of less than no observable time and still be ordered with respect to the positive entropy arrow of time. Also note that the measure of time in a negative time, positive entropy inner region can be governed by a negative metric such that a zero or null amount of positive time corresponds to an infinite amount of negative time i.e., a region of spacetime in which an interval of negative 10 temporal units is smaller than an interval of negative 1 temporal units, which in turn is smaller than the largest negative time interval which is zero temporal units.

Although observers in an outer region of spacetime cannot directly observe and thus scientifically verify what is taking places within the null inner regions of the universe, there is a wealth of indirect evidence to support the conclusion that these virtual inner regions of the universe are active and that what is taking place in them can affect what is taking place in the outer observable universe (see Casmir effect, Lamb shift, Vacuum Birefringence, . . .).

With respect to modern relativistic theory these null interior regions are depicted as the spacetime regions bounded by the luminal null geodesics that represent the shortest path of influence that past manifestations of IR potential energy can exert on the present location of an elementary particle in spacetime and the null geodesics that connect the influence that a presently located elementary particle can exert on future manifestations of IR potential energy in the observable universe.

The geodesics of other elementary particles that exist in interior regions depicted by the horizontal super luminal cone in figure 2 represent a determined inner region of spacetime that cannot be changed by an observer located at the nexus of the horizontal virtual light cone. Note again, that relative to observers in outer regions all transitions that particles within these inner regions undergo will appear to take place instantaneously, but this fact does not exclude the possibility that events in these inner regions could be taking place in imaginary intervals of time or in less than no time negative temporal intervals. Note also that when the retarded position of an elementary particle is very close to an observer the determined segment of its virtual existence will graphically appear to be very small which implies that as the spacetime distance between elementary particles decreases fluctuations in the IR energy, momentum and spacetime position of elementary particles due to the Heisenberg uncertainty principle will become significant and when the retarded position of an elementary particle appears to be very far away from an observer the determined segment of its virtual existence will graphically appear to be very large and the influence of the Heisenberg uncertainty relationship on the IR energy, momentum and spacetime position of elementary particles will become insignificant. The unobservable states and transitions of the unobservable inner space manifestations of elementary particles will be represented by IR world line segments that will be called *filaments*.

The traditional relativistic depiction of spacetime events becomes problematic when attempting to model events in term of quantum probabilities because in quantum field theory elementary particles are represented by knots of entangled matter waves that have super luminal phase relationships and either real luminal (Bosons) or real sub luminal (Fermions) group velocities and are graphically depicted with respect to a probability space that does not contain a temporal axis —Are mathematically treated using complex (Argandian) mathematics in which

the coordinates and properties of elementary particles are expressed in terms of the sum of real and imaginary numbers, that must be mathematically manipulated in order to give meaningful real answers as well as solutions that must be ignored or discarded because they appear to be meaningless. Given that these two ways of depicting particle interactions are so different, it is no wonder that the unification of these two very accurate and precise theories has stymied theoretical physicists.

The traditional use of an imaginary dimension of time to map spacetime coordinates in relativistic theory is strange because if time is represented as an imaginary dimension with coordinates expressed in spatial units of *ict,* then according to the rules of traditional complex analysis it is not possible to mathematically distinguish between +i and -i units on the temporal axis, and thus there is no way to scientifically say that 2i units of time is less than 4i units of time, which implies that there is no way to distinguish between positive or negative imaginary time intervals, which has implications for quantum field theories constructed on the assumption that elementary particles and their antiparticles move in opposite directions relative to an imaginary physical dimension of time —An assumption born out of reading too much into graphical depictions of spacetime events that use a linear or curvilinear temporal coordinate axis to represent something that is measured by a device that goes around in a circle or oscillates with a constant period (that go nowhere).

The representation of elementary particles moving with respect to an imaginary dimension of time is also strange because the motion of elementary particles in time could only be observed by an observer located at the origin of spacetime and since no such frame of reference exits in the observable universe, the dimension of time is a fictitious dimension and the representation of the motion of elementary particles that are comoving with respect to the

dimension of time violates the principle of relativity, i.e., how could elementary particles moving with equal velocity in the same direction of spacetime determine that they are in motion in the absence of any external reference points.

If it is assumed that the arrow of time with respect to the transfer of information in the observable outer region of the universe points in a positive entropy direction (from less positive to more positive for spontaneous processes), then the sum effect of all energy/information transferring, time delaying, phase altering of signals originating in the past and arriving at a point of convergence (the points P_1 and P_2 in figure 2) can be represented by a fictitious convergent time forward positive entropy IR energy wave that has both real and imaginary components, i.e., as a convergent (deflating) time forward positive entropy IR energy wave comprised of real and imaginary components the intensity (energy density) of which appears to go from zero to infinity as $(c\tau)^2$ goes from negative infinity to zero relative to observers located in the observable universe that are comoving with P_1 and P_2, and all components of an elementary particle's IR convergent energy wave that interact with P_1 and P_2, and are redirected by P_1 and P_2 towards a future state of the universe can be represented by a divergent (inflating) fictitious time forward positive entropy IR energy wave comprised of both real and imaginary components the intensity of which decreases from infinity to zero as $(c\tau)^2$ goes from zero to infinity relative to observers comoving with P_1 and P_2. Note, that in order for the photons that make up light waves to appear to move with wave like motion at the speed of light relative to observers in an outer region of spacetime and not violate the uncertainty principle, the principle of relativity and violate causality by appearing at times to move backwards with respect to the dimension of measured time, requires that the photons appear to oscillate between moving faster than an observer's rate of motion and slower than an observer's rate of motion with respect to the fictitious dimension of

time. That is, if we assume that an elementary particle is moving through the dimension of time at a rate of v, then the Heisenberg uncertainty inherent in the measure of its velocity must be reported in units of $\pm\sigma$, which implies that sometimes the elementary particle will be measured to be moving faster and sometimes slower through time and this variance will mean that an elementary particle will sometimes appear to an observer to be moving forward and sometimes appear to be moving backwards with respect to the dimension of time. What needs to be kept in mind here is that although the photon may at times appear to be moving backwards through time it is actually moving forward through time at a rate of $v-\sigma$ with respect to the rate at which an observer is moving with respect to the fictitious dimension of time. This means that if an observer were to attempt to very precisely measure the location of a co-moving elementary particle in time, that elementary particle would appear to have a spectrum of possible IR energies with respect to each dimension of spacetime that are related to the doppler shifting of the signals that must either be emitted by or scattered or reflected off the elementary particle that is being observed, i.e., an observed comoving elementary particles should appear to undulate in spacetime like a cork bobbing up and down on the surface of the ocean.

This also implies that every measurable point of space-time acts as an undulating convergent sink and an undulating divergent source of IR energy and since in quantum theory the measure of energy and time, and momentum and distance are conjugate observables, the distribution of real and imaginary (IR) energy in the observable regions of spacetime will appear to be quantized, and the standard deviation of the mean measure of how much IR energy is contained in a measurable volume of spacetime will depend on how precisely one attempts to measure the intervals of spacetime used to determine the mean. Ultimately this implies that mathematical models based upon observations made under terrestrial conditions will fail to

accurately predict the behavior of concentrations of IR potential energy separated by both relatively large and small real and imaginary distances and that it should not be possible to precisely determine the momentum and position of the center of IR potential energy of a large ensemble of elementary particles by simply summing the measurements made on the individual elementary particles that make up the ensemble.

Early in the history of quantum theory it was thought that elementary particles existed in states of superposition until the act of measurement disrupted the state of superposition and collapsed their wave functions.[6] This notion that particles do not exist until we measure their state properties harkens back to the old philosophical debate over whether or not a tree falls in a forest if no one is there to observe it. The main problem with this belief is that it does not sit well with quantum field theory in which virtual fields are interacting with elementary particles in unmeasurably small intervals of spacetime, which implies that the fields that give rise to the appearance of elementary particles must be able to measure the non-commuting conjugate variables of all the elementary particles in spacetime to such a degree of precision that deviations around the mean of the non-commuting conjugate variables is so small that human beings cannot measure the degree to which they are fluctuating without great difficulty and the use of sophisticated technology but not to such a small degree of precision that the ability of a particle to measure one of the non-commuting conjugate variables interferes with its ability to precisely measures the other, i.e., if the fields that give rise to elementary particles could measure the spacetime position of the other elementary particles in the observable universe to an infinite degree of precision, then the fields would "know" nothing about the momentum/energy of the elementary particles in it, which would result in chaos. The realization that the interaction between particles and fields cannot violate Heisenberg uncertainty principle implies that the rate

at which elementary particles appear to accelerate in response changes in the fields that mediate their interactions must vary but not so much as to make those variations easy for us to discern, i.e., the precision with which the gravitational field of a star can determine the position of an orbiting planet or the gravitational field of a planet can determine the position of a free falling object must be greater than zero and also greater than the degree of precision that would allow scientists to easily observe variation in the rate at which planets orbit stars and objects appear to free fall towards the surface a planet. Ultimately what this means is that debates over the role that human beings play in determining where and when elementary particles exist or whether or not trees fall in a forest when no one is there to observe them fall, are for the most part irrelevant and that the observable universe will be able to get along fine without us, i.e., will not cease to exists when we are no longer there to observe it.

Although the conjugate properties of position and momentum and energy and the location of an elementary particle in time are often represented as distinct things in introductory text on quantum mechanics,[2] the conclusions of modern relativistic theory, specifically the interdependence of spatial and temporal measurements and the fact that mass is proportional to energy requires that they be treated as one unified uncertainty relationship.

$$\sigma x \sigma p \geq \frac{h}{2\pi} =$$

$$\sigma x \sigma (mv) = \sigma x \sigma \left(\frac{E}{c^2}\right)\frac{\sigma x}{\sigma t} = \sigma x \frac{\sigma E}{\frac{\sigma x^2}{\sigma t^2}}\frac{\sigma x}{\sigma t} = \sigma E \sigma t$$

$$\therefore \frac{\sigma x \sigma p}{\sigma E \sigma t} = 1 \qquad [1]$$

The understanding that every observable volume of spacetime contains both real and imaginary forms of potential energy and that the amount of IR potential energy that is measured in a volume of spacetime is dependent upon the Heisenberg uncertainty relationships, leads to

the realization that a classical inverse square law like Newton's law of Gravity will fail to predict the behavior of two masses when the distance separating them is relatively large because if every point of spacetime has a potential energy associated with it and as Isaac Newton demonstrated, the gravitational force acting between two elementary particles/objects is dependent upon the potential energy contained in a measured volume of spacetime defined by a radius equal to the distance separating a source and an elementary particle or collection of elementary particles, then Newton's law of gravity will need to be adapted as follows to include all the potential energy in that spherical region of spacetime.

$$F_G = G\frac{(E_1 + \rho V)E_2}{c^4R^2} \qquad [2]$$

In which the ρV term represents the energy contained in the space enclosed by the volume of a sphere with a radius R extending from an object E_1 to an object E_2 excluding the energy contained within the two objects. Note, when R is close to the radius of a macroscopic source the ρv term can be ignored and equation [2] reduces to Newton's law of gravity.

$$F_G = G\frac{(E_1 + \rho V)E_2}{c^4R^2} \approx G\frac{(E_1)(E_2)}{c^4R^2} \qquad [3]$$

Also note that Einstein based the derivation of the field equations of the General Theory of Relativity on the assumption that their predictions should correspond to the predictions of classical gravitational theory in the weak field limit and the Schwarzschild solution of the field equations of General Relativity were based upon the assumption that the space between masses was empty,[7] which implies that the General Theory of Relativity as originally formulated is conceptually flawed and will catastrophically fail to predict the behavior of matter when the distance separating a source and a test mass is relatively large, i.e., fail to predict such things as the orbital speeds of stars in galaxies and the correct angle that light is deflected by very large

systems such as galaxies and clusters of galaxies and the relationship between the cosmological red shift and the inflation rate of the observable universe. Ironically it is this oversight that has led many to erroneously hypothesize that the space between stars and galaxies is filled with copious unobservable quantities of dark matter and dark energy, and to arrive at their estimates of the amount of dark matter and dark energy in the observable universe by comparing the observed behavior of the macroscopic universe to the erroneous values predicted by ill-conceived classical and relativistic gravitational theory.[7]

Examining the Minkowski light cone commonly employed in the special theory of relativity (figure 2) it is evident that the components of the convergent IR energy waves that manifest themselves as elementary particles could be affected by the divergent IR energy waves emanating from the retarded position of other elementary particles in the universe and that the mixing of the convergent IR energy waves of one elementary particle with the divergent IR energy waves of another elementary particle could alter the most probable position of a particle in spacetime which would give rise to asymmetries in an elementary particle's convergent IR energy wave and too the perception that elementary particles were being motivated to move relative to one another by some kind of mysterious or magical force.

In quantum theory the smallest unit of measurable distance between events in the observable universe is a Plank length of spacetime and all events that occur in less than a Plank length of time occur in what appears to an observer in an outer region of spacetime to be no time at all, and indirect evidence must be used to support the hypothesis that the observable properties of elementary particles are being affected by their interaction with null or quanta of virtual IR energy contained in unobservable inner regions of the universe. This implies that the presence of active inner spaces should give rise to what appears to observers in the outer observable universe

to be mysterious or magical ubiquitous forces, with the word magical being used in the sleight of hand sense i.e., in the sense that "the hand is faster than the eye."

Since it would be unrealistic to assume that the quanta of the past to future IR energy waves that connect past states of the observable universe to the present position of an elementary particle arrive simultaneously and symmetrically at the nexus of any elementary particle in it, it is expected that at the plank scale all elementary particles should exhibit IR chaotic Brownian motion which will give rise to observable random Brownian motions in the microcosm that will be smoothed away in large ensembles of elementary particles.

$$\langle \boldsymbol{p} \rangle_{re} = \sum_{m} \langle p \rangle_{(re)m} \approx 0$$

$$\langle \boldsymbol{p} \rangle_{im} = \sum_{n} \langle p \rangle_{(im)n} \approx 0 \qquad [4]$$

In which the $\sum_m \langle p \rangle_{(re)m}$ represents the sum of all interaction of the real components of boundary quanta (contains the contributions of both real and virtual real convergent and divergent interactions) and $\sum_n \langle p \rangle_{(im)n}$ represents the sum of all interactions of the imaginary components of boundary quanta (contains the contributions of both imaginary and virtual imaginary divergent and convergent interactions). It is important to note that any asymmetries in the distribution of potential energy in the retarded real, the imaginary, the virtual real and the virtual imaginary regions of spacetime will give rise to asymmetries in a particle's Brownian momentum which with respect to the virtual components of a particle's Brownian momentum will give rise to the illusion that mysterious or magical forces exist in the observable universe.

The fact that the retarded arrangement of IR potential energy about any elementary particle is unique implies that the boundary of each elementary particle is unique, which in turn implies that each elementary particle is a unique expression of IR energy waves, that each

elementary particle exists in its own unique observable universe, that in the strictest sense there are no such thing as indistinguishable particles, and that elementary particles can never be at rest relative to each other, which precludes the existence of rest frames of reference.

Since quantum theory predicts that the ground state energy of an elementary particle in a potential well cannot be zero (Heisenberg uncertainty principle)[2] then anything that increases or decreases the uncertainty in the probability that an elementary particle is moving in a particular direction relative to another particle, or produces an asymmetry in the direction or magnitude of the IR energy imparted to an elementary particle by other elementary particles, or that could create asymmetries in the phase of the IR energy waves that arrive from the origin of the universe and manifest themselves as elementary particles, will give rise to the appearance of accelerated motion. Thus, in order to develop a coherent quantum based relativistic mathematical model that can be used to predict the motion of large aggregates of elementary particles in the observable universe, all possible asymmetries in the retarded IR potential energy density of the spacetime that appears to surround an elementary particle or aggregates of elementary particles will need to be examined with respect to the type of effects that such asymmetries have on the behavior of elementary particles in the observable universe.

The converging IR energy waves that arrive at the nexus of each elementary particle in the observable universe need to be analyzed with respect to three types of interactions:

1) Energy transferring interactions, i.e., a bowling ball striking a bowling pin, or a gas molecule bouncing off a grain of pollen), which can be modeled using classical and relativistic collision theory), or as an energy exchange between an elementary particle and an asymmetric field of bosons, or as an asymmetric change in the strength that an elementary particle couples to the Higgs field.

2) Phase induced changes in an elementary particle's quantum observables i.e., energy, momentum, and position.
3) Boundary asymmetries induced by asymmetric distributions of potential energy around elementary particles, which will make the observable universe appear to be older/colder/bigger/less dense/more red shifted/more entropic in a particular direction, and younger/hotter/smaller/more dense/more blue shifted/ less entropic in the opposite direction.

Statistically speaking, the large numbers of interactions that must be accounted for when describing field-particle interactions implies that the mathematical models used to describe field-particle interactions need to be constructed upon the understanding that large numbers of symmetric field-particle interactions will give rise to Gaussian solutions, and the concept of skewness (curvature) which is used to gauge the amount of asymmetry in a system by comparing the skewness of an asymmetric probability distribution to the ideal 0 skewness of a pure Gaussian probability distribution.

In the depiction of the Minkowski light cone (figure 2) that is commonly used in the Special Theory of Relativity, the nexus of the IR energy waves that appear to be converging towards and diverging away from elementary particles in the observable universe are represented by the points P_1 and P_2, which coincide with the fictitious center of mass/charge of an elementary particle or ensemble of elementary particles. In this depiction the quanta or elementary particles that make up the converging energy wave will either be transmitted through, absorbed and re-emitted by, reflected or scattered off P_1 and P_2 towards a future state of the observable universe. This implies that asymmetries in the macroscopic vector sum of all expectation value altering components of an elementary particles' converging IR energy wave will alter the momentum of

that particle relative to the other elementary particles in the observable universe. This also implies that all elementary particles are both a sink and a source of quantized IR energy waves.

A Modified (IR) Model of Spacetime.

In this relativistic quantum mathematical model of spacetime, proper time τ will be represented using real irrational temporal coordinates usually expressed in units of seconds, minutes, hours, . . . or by standardized irrational dimensionless multiples of π (time represented as something that goes around in circles with a constant angular velocity or moves harmonically with a constant period).

In this model and unlike previous quantum or relativistic models, the relative position of elementary particles in space will for this introduction be represented using imaginary and real (IR) spatial coordinates mapped with respect to linear orthogonal real (X_{re}) and imaginary (X_{im}) spatial dimensions.

In this IR model of spacetime, the principal momentum vector or PM vector of an isolated elementary particle in the observable universe will be defined to lie in the IR trace plane along the real X_{re} axis (see figure 3).

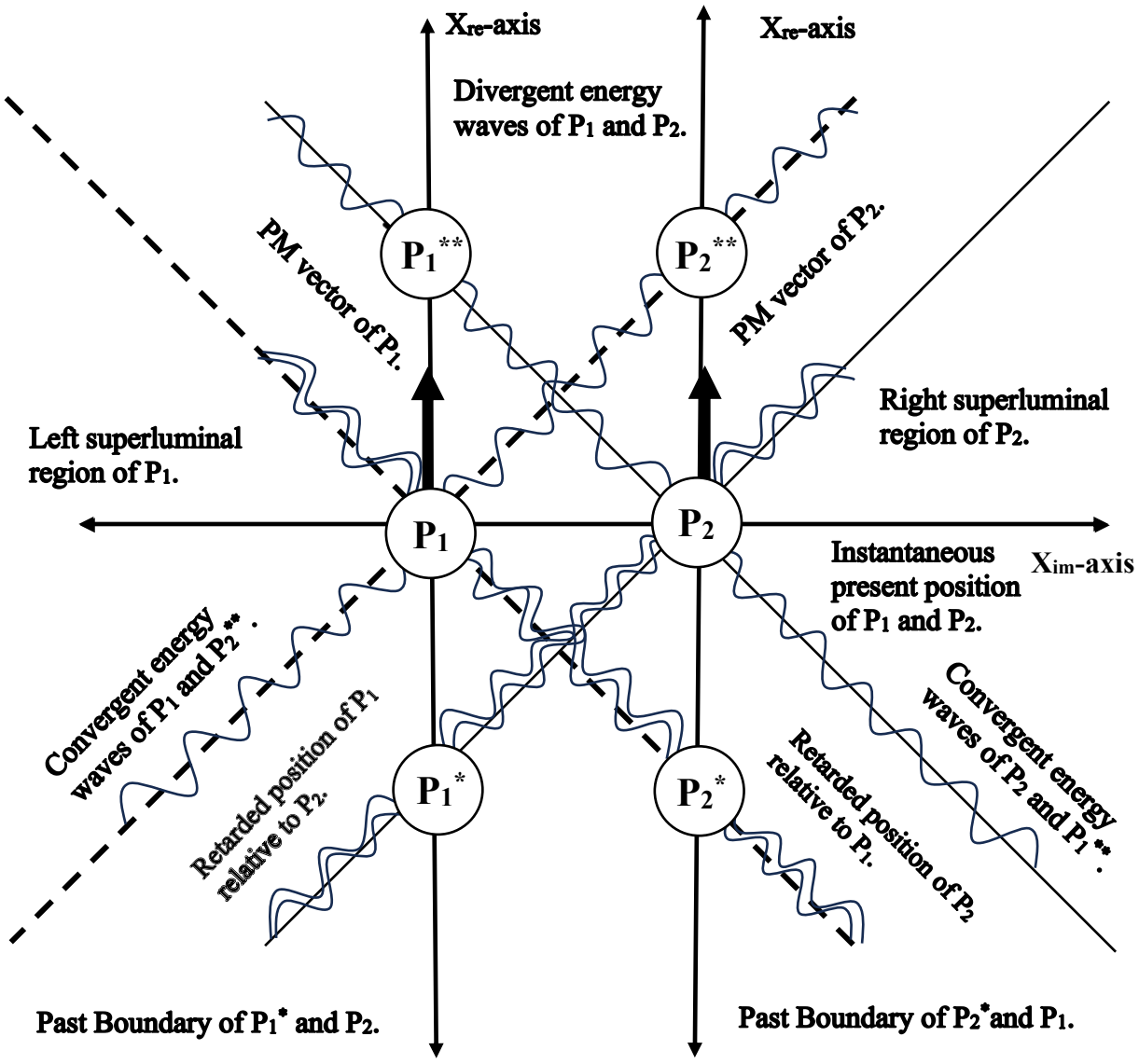


Figure 3

Fictitious IR Frame of reference representing the position of two comoving particles P_1 and P_2 in the IR spatial trace plane.

Note that the real spatial coordinate axis is a mathematical construct and an elementary particle would only appear to be propelled along its' (X_{re}) axis relative to an observer that is at rest with respect to the origin of spacetime and since no such frame of reference actually exists in the

observable universe, this frame of reference will be defined to be a fictitious frame of reference. In this model all comoving elementary particles will be said to have a real fictitious IR momentum of zero associated with them.

Ignoring for now the very real possibility that elementary particles can tunnel across the boundaries that separate inner and outer regions of spacetime, the IR energy waves that travel from the past to the present position of an elementary particle will be defined to consist of 1) real observable elementary components, 2) real virtual elementary components, 3) luminal boundary elementary components, and 4) super luminal imaginary virtual components, and that these four converging energy wave containing regions of IR spacetime can be broken down into at least four distinct energy spectrums classified as follows.

1) An energy/momentum spectrum consisting of converging (past to present) IR sub luminal elementary particles that move with various IR Kinetic Energies within the region of a relativistic light cone that is mathematically defined to be an observable universe or the outer region of the universe in which events are separated by real spacetime distances.
2) An energy/momentum spectrum consisting of converging (past to present) IR sub luminal virtual elementary particles (fermions) that move with various Kinetic Energies within the region of a relativistic light cone that are mathematically defined to be “real virtual null regions,” i.e., to exist in regions in which events are separated by real spacetime distances but because of technological limitations appear to be separated by null distances or to contain unmeasurably small quantities of IR energy.

3) An energy/momentum spectrum consisting of converging (ever present) luminal boundary elementary particles (Bosons) that move with various Kinetic Energies on the boundary of observable spacetime, i.e., observable bosons such as photons.
4) An energy/momentum spectrum consisting of converging (ever present) IR super luminal null virtual elementary particles that move with various Kinetic Energies within the regions of a relativistic light cone that are mathematically defined to be "imaginary inner spaces".

In this IR model of spacetime an ordered inner imaginary fictitious coordinate axis X_{im} will be used to connect the virtual contemporary instantaneous position of all the elementary particles in a comoving group (see figure 3) and the time static (null) instantaneous trace IR spatial distance between any two events in the IR trace plane will be determined by using the following pseudo Euclidean non-Argandian metric.

$$d = \sqrt{\alpha(\Delta x_{re})^2 + \beta(\Delta x_{im})^2} \qquad [5]$$

In which α and β are weighted statistical operators that will allow for the range of possible outcomes to be predicted in such a way that the certainty one can determine the outcome of a measurement increases with the number of elementary particles in an ensemble of elementary particles. Note that this IR metric allows for the macroscopic spatial distance between any two events in this IR spatial trace plane to be represented as being separated by either null, real, or imaginary static, expanding or contracting distances and in figure (3) real trace geodesics are defined to exist when $|(\alpha\Delta x_{re})^2| > |(\beta\Delta x_{im})^2|$, imaginary trace geodesics are defined to exist when $|(\alpha\Delta x_{re})^2| < |(\beta\Delta x_{im})^2|$, and null trace geodesics are defined to exist when $|(\alpha\Delta x_{re})^2| = |(\beta\Delta x_{im})^2|$.

If one includes a third fictitious dimension to represent τ (an observer's proper measure of real positive entropy time) then a third real fictitious coordinate axis labeled $c\tau$ can be employed, in which τ_{re} represents real intervals of time measured by an observer and the IR metric for this three-dimensional slice of spacetime is given by

$$d = \sqrt{\alpha(\Delta x_{re})^2 + \beta(\Delta x_{im})^2 + \gamma(\Delta \tau_{re})^2} \qquad [6]$$

In which α, β, and γ are operators that must be set to values that correspond to the observable properties of events separated by the spacetime distance d. This model could be extended to include n dimensions, but the minimum number of dimensions needed to model events in a real valued observable universe is 7, three real spatial dimensions, three imaginary spatial dimensions and one real positive entropy dimension of time.

Assuming that the two particles P_1 and P_2 depicted in figure 3 represent large ensembles of elementary particles that appear to be comoving with respect to an observer 'O', then any change in the velocity of P_1 or P_2 can be represented by a boost or rotation of the fictious PM vector of P_1 or P_2 relative to that observer. If relative to an observer located at the point P_1 or comoving with P_1, P_2 appears to be moving with constant velocity v, then a boost angle $\pm\theta$ can be used to represent the orientation of the PM vector of P_2 relative to the PM vector of P_1. If it is assumed that the velocity of an elementary particle equals the velocity of light when the PM vector of an elementary particle is orientated at an angle of ±45° relative to the PM vector of another elementary particle, then the apparent velocity (momentum/kinetic energy) of P_2 relative to another observer could be naively represented by the equation $v=c(\tan\theta)$ in which c is the velocity of light, θ represents the angle between the PM vector of P_1 and P_2, with the angle between them being assigned a value of zero when P_1 and P_2 appear to be comoving or at rest with respect to each other and v represents the velocity of P_2 relative to an observer co moving

with P_1. This naïve depiction represents a pseudo classical conceptualization of velocity i.e., a fictitious universe in which the relative velocity of elementary particles would appear to range from zero to infinity. Because we know that the velocity of light represents the limit at which information can be transmitted from one point to another in the observable universe, and that the relationship between a force acting on an elementary particle and the motion it produces is hyperbolic, we know that this classical view of space and time can only be used to accurately model the behavior of elementary particles when the relative velocity of the elementary particles is much less than the velocity of light. Because of length consideration no attempt will be made here to rederive the already well-established relativistic transformation laws as put forth in the special and general theories of relativity and relativistic quantum theory. However, as in relativistic theory, it will be assumed in this IR model of spacetime that elementary particles that appear to accelerate from some initial velocity v_i to some final velocity v_f will also need to follow non-linear trajectories. Note, simply setting $\alpha = 1$ and $\beta = \frac{v^2}{c^2}$ in equation 6 would go a long way towards modeling length contractions (deflation) and using a velocity dependent inflationary operator γ to model time dilation (time inflation) would go a long way towards reproducing the Lorentz transformations of special relativity, in which spatial contraction and time inflation are harmonically linked to each other. However, as previously noted, the use of a fictitious coordinate axis to represent the measure of time is a conceptually flawed artifact that was carried over from classical theory into modern relativistic theory and as such will be of little use in the construction of this quantized IR model of spacetime.

In this IR model of spacetime, non-linear trajectories or geodesics are the result of the harmonically linked inflation and deflation of the real and imaginary dimensions of spacetime in which the rest distance between two comoving points is imaginary (virtual), the luminal distance

between points is zero and the luminal geodesics separate real outer spaces from imaginary inner spaces.

If it is assumed that the measure of time is real in real regions of spacetime, and imaginary in imaginary regions of spacetime, then the IR metric used to define the spacetime distance between macroscopic events in the IR observable universe would need to be updated to the much more complicated

$$d = \sqrt{\alpha(\Delta x_{re})^2 + \beta(\Delta x_{im})^2 + \gamma(\Delta \tau_{re})^2 + \ \delta(\Delta \tau_{im})^2} \qquad [7]$$

This equation implies that the relationship between α,β,γ,δ will have to be chosen so as to reproduce the Lorentz transformations of special relativity for local observable regions of the universe in which the spatial trace of events is defined to be real when $[(\alpha x_{re})^2 + (\gamma \tau_{re})^2] > |[(\beta x_{im})^2 + (\delta \tau_{im})^2]|$. Of course, the α,β,γ,δ operators need to be chosen so as to model the actual behavior of elementary particles in the observable universe in which the measure of spatial and temporal intervals can be either directly proportional to each other, or harmonically or an-harmonically linked. Before the relationship between these operators can be determined with respect to a fictitious frame of reference, i.e., how everything moves relative to the origin of space-time, a much deeper understanding of how observers determine the velocity of objects and the distance between objects in the observable universe will need to be clarified.

Finaly it is important to note that in the IR timeless trace plane the angles between any vector that connects the virtual present position of a moving object to a future position of that object in the IR trace can be real or imaginary.

Relativity Revisited

Classically the velocity of an object is naively defined to be,

$$\frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \qquad [8]$$

i.e., as the change in the distance an object appears to move, divided by the amount of time that is observed to pass as the object moves from a point x_1 to a point x_2.[9] This definition of velocity was found to be flawed when it was discovered by Einstein that the simultaneity of two events, such as the act of simultaneously measuring the two ends of a ruler, is a relative property of spacetime.

Einstein developed a coherent relativistic definition of velocity by using the fact that the speed of light is an invariant property of spacetime in conjunction with the understanding that the simultaneity of two events was not an invariant property of spacetime. In order to create a mathematical model that was consistent with observation, Einstein defined the distance between two points P_1 and P_2 to be the speed of light (c), multiplied by half the time it takes a light signal to travel from a point P_1 to point P_2, and back again to P_1, i.e.,

$$d = c\left(\frac{\Delta t}{2}\right). \qquad [9]$$

Einstein then used this definition of distance as the basis for synchronizing the clocks of observers.[10.] Unfortunately, this definition of velocity and the Special Theory of Relativity it spawned were created before the discovery of the CMB, and with its discovery the realization that all observers can determine their velocity relative to the velocity of other observers by measuring asymmetries in the temperature of this nearly isotropic, ubiquitous universal event.

Rethinking Relativity.

Let two observers A and B appear to be at rest with respect to each other, and far removed from any significant sources of IR potential energy, and for sake of simplicity let it also be assumed that the CMB appears to be symmetric relative to both observers. If a fictitious IR trace plane is used to represent the trajectory of A and B relative to each other, then the PM vectors of A and B will be parallel to each other, which relative to observers at A and B implies that motion

along these real inflation vectors is not an observable property of the spacetime continuum — That A and B will appear to be at “rest” with respect to each other.

Let A and B each use a laser to determine the distance between them in the usual way, i.e., by measuring the time it takes a light signal to make the round trip between A and B and then dividing that quantity of time by two and then multiplying it by the speed of light. Let A and B define that distance to be $c\tau_A$ and $c\tau_B$ respectively, and both agree that $\tau_A = \tau_B = 1$ light minute. Let the observers at A and B agree to synchronize their clocks by letting observer A at precisely 11:50 AM send a radio signal to a clock located at B that upon receiving the signal from A will automatically be set to 11:51AM the “instant” it is received. Let A and B agree that at precisely 11:59 AM ‘A’ will send out a signal at the velocity of light that will upon being received by B precisely at 12:00 PM relative to B, launch a collection of 10 fantastical rockets each carrying a fantastical observer and a fantastical atomic clock from B to A with velocities that range from .1c up to a velocity so close to the velocity of light that no known measuring device could be used to determine a difference between its velocity and the velocity of light. In using the word fantastical here, it is being assumed that the rockets can instantly accelerate and decelerate to their respective initial and final velocities. It will also be assumed that the distance between A and B will be defined to be equal to $c\tau_A$, the distance as measured by observers that are comoving with respect to A and B and that no attempt will be made on the part of the fantastical rocket observers to measure or attempt to determine the instantaneous distance between themselves and A or B while they are in transit or to look for asymmetries in the macroscopic universe such as stars appearing to be blue shifted in a particular direction and red shifted in the opposite direction or to look for temperature asymmetries in the CBM or

temperature differences in the hull of the space ships they are travelling in due to space friction (i.e., it is being assumed here that space is a perfect vacuum).

Upon arriving at A, each observer that moved from B to A will compare the time they measured to make the trip from B to A to the time it took them to make the trip as measured by those at rest with respect to A and B, and then use their elapsed time to infer the velocity they were traveling at. It is expected that at the end of this experiment the observers that were at rest with respect to A and B will conclude that the velocity of the fictitious rockets ranged from .1c to c, but the fictitious observers who made the trip will conclude that they were in fact traveling at velocities that ranged from a little faster than .1c to an infinite velocity. Thus, we see that how we define velocity determines whether we believe that we live in a universe in which the velocity of light is a finite velocity, or the velocity of light is an infinite velocity. Which brings us to the possibility that one of these two very different ways of defining velocity may actually have physical implications about the nature of the observable universe.

Although removing the above observational restrictions and allowing each of the moving fictitious observers to measure the distance between A and B while they are in motion will necessitate the use of hyperbolic mathematics to transform the observed spacetime coordinates between successive events from one frame of reference to another, it does not alter the conclusion that only in the trivial mathematical sense can the observers in the rockets claim that it does not matter if they assume that they are moving towards A and away from B or if they assume that A is moving towards them and B is moving away from them, because once the observational restriction are removed and the observers in the rocket ships are allowed to observe the entire universe they will find all kinds of motion induced asymmetries in the structure of

spacetime that will enable them to distinguish between whether A is moving towards them or if they are moving towards A.

For example, if a car A is moving along a highway at 50 MPH, and is passed by a car B moving 70 MPH, kinematically speaking it is ok to say that relative to an observer moving along with B, that A is receding from B at 20 MPH, but this observation is only true in a superficial sense because if an observer moving along with B looks at the wheels of A, they will see that the wheels of A are spinning in a direction that precludes the possibility that A is moving away from B. Further, observers traveling in car B could also measure the temperature of the front and back of the car, and find that the air in front of the car is warmer than the rear of the car due to the compression of air in the direction of motion and that the temperature of car B is not the same as the temperature of car A. In fact, this temperature difference will be found to exist even in outer space, because spacetime is not a perfect vacuum, and the average density of spacetime is not an invariant property of spacetime i.e., as the velocity of an object goes to the speed of light, the density, pressure, and temperature of spacetime will appear go to infinity in the direction an object is moving, and zero in the opposite direction.

As in the theory of relativity, when elementary particles appear to be at rest with respect to each other they will be called comoving elementary particles, and a collection of comoving elementary particles will be defined to be a comoving group. It is important to note that the PM vector of elementary particles with similar asymmetries in their past can be represented as being rotated relative to those with a different degree of skewness or none at all. Thus, in this new IR model of spacetime, all relative motion can be represented by the relative orientation of the fictitious PM vectors of the elementary particles being examined.

It is expected that the introduction of time dependent skewness changing asymmetries in

the IR potential energy density of the retarded space that appears to surround an elementary particle will give rise to the appearance of accelerated motion i.e., to a dynamically evolving observable universe in which elementary particles appear to be evolving in response to mysterious or magical forces.

The Mass Charge Energy Conundrum.

Since the classical era, theoretical physicists have been building mathematical models around the notion that matter consist of particles that contain a potential form of energy called mass and a potential form of energy called electric charge. One of the great modeling tools that mathematical physicists have employed to construct very precise models of systems comprised of elementary particles that contain mass and electric charge is the superposition principle, which allows physicists to very precisely model and predict the behavior of large macroscopic ensembles of mass and electric charge by summing the mass and electric charge containing components of macroscopic systems into localized fictitious points called the center of mass, and the center of electric charge. Although mathematical models that utilize superposition simplifications are very precise, and the mathematical equations utilized in these models are relatively easy to solve when compared to the daunting task of trying to arrive at solutions to many body problems, the mathematical models they give rise to are mute with respect to two very important scientific and philosophical questions:

1) Can the center of the electric charge in atomic systems be represented as residing at a single point in atomic system if the Pauli exclusion principle and the Heisenberg uncertainty principle requires that negatively charged electrons reside outside the quark confining nucleons contained in the nucleus?

2) At what degree of precision would the superposition of all the electric charges contained in charge symmetric systems, i.e., systems that contain equal numbers of positive and negative electric charges, no longer appear to be electrically neutral?

The Pauli exclusion principle and the Heisenberg uncertainty relationships require that the negative electric charge of the electrons in atoms exists in orbital regions that reside outside the nucleus of the atoms, i.e., are forbidden to exists within the quark containing nucleons that reside in the atomic nucleus; a restriction that implies that at the quantum level, the electric charges contained in large macroscopic ensembles comprised of large numbers of atomic systems need to be represented by at least two distinct moving non-superimposable fictitious centers of electric charge, a positively charged fictitious center that is relatively massive and static and that represents the sum of all the electrically charged quarks contained in all the atomic nuclei in the ensemble and a negatively charged, much less massive and often times asymmetrically oscillating fictitious center of negative electric charge that represents the instantaneous sum of all the negatively charged electrons in the ensemble; which when taken in conjunction with asymmetries inherent in inverse square relationships that are related to the fact that the electrons in a particular atomic system are sometimes closer to and sometimes farther away from the electrons in another atomic system, while the variation of the distance between the quarks in the nucleons is relatively small by comparison, implies the existence of an ubiquitous fundamental overall attractive oscillating electric force replete with a magnetic component and an overall attractive time varying dipole/dipole force acting between and within the components of all the atomic systems in the observable universe.

To clarify what is being hypothesized here, let us examine the assumptions that are employed in the calculation of the Newtonian gravitational force acting between two masses

such as the earth and the moon. Traditionally, the assumption is made that all the mass in the moon and all the mass in the earth exists in two distinct fictitious centers of mass, and that all the electric charges in the earth and the moon sum to zero, which reduces the task of calculating the magnitude of the gravitational force acting between the moon and the earth to using Newton's law of gravity i.e., precisely measuring all the mass in the earth and the moon, and precisely measuring the distance from the center of mass of the earth, to the center of mass of the moon. The problem with this is that the Pauli exclusion principle and the Heisenberg uncertainty relationships place restrictions on where electrons in macroscopic systems comprised of many atomic systems can reside relative to the quark containing atomic nuclei in those systems, which implies that at best the superposition of the electric charges and masses contained in macroscopic ensembles of matter can be represented by a quantized fictitious hydrogen atom in which a fictitious proton that contains the mass and electric charge of the all the quark containing protons and neutrons in the ensemble of atomic systems is surrounded by a fictitious electron that contains the combined mass and electric charge of all the electrons in the ensemble and because the inverse square law requires that when the distance separating the electrons in one atom from the electrons in another atom is a minimum (both electrons reside between the atomic nuclei) the sum of all the electric forces acting between the atomic systems will result in a maximum repulsive force which, as Richard Feynman demonstrated in his important 1939 paper, will give rise to an attractive dipole-dipole force between the atoms.[11]

In a paper published in 2015 it was reported that Mone Carlo computer simulations of the net electric force that arises when the electrons in the s orbitals of two H_1 hydrogen atoms separated by a distance of one meter randomly exchange virtual photons with each other and the protons in each H_1 atom revealed the existence a small net attractive residual electric force that

was comparable in magnitude to the gravitational force acting between the two H_1 atoms.[12] Further it was found that for certain arrangements of quarks in nucleons that the electric van der Wall forces would become extremely large when the distance between nucleons was on the order of 10^{-15} m. Although these controversial and rather surprising computer simulation-based findings were published they did not elicit a response from the scientific community and until they are either confirmed or refuted it will be conservatively assumed here that no relationship exists between the most probable arrangements of the electric charges in atomic systems and the magnitude of the gravitational force. Thus, in this model no attempt will be made to unify these two important physical properties of elementary particles and the mathematical modeling of the effect of electric charge and gravitational mass will be carried out with respect to a bifurcated but localized IR entity called the IR potential energy, with the words mass and electric charge being used to distinguish between the imaginary and real components of IR potential energy respectively.

Since at this time there is no scientific evidence to support the hypothesis that the gravitational force is a manifestation of the electric force, it will be assumed that the modeling of the relative motion of elementary particles in the observable universe needs to be derived from the interaction of the convergent IR energy waves that manifest themselves as elementary particles and that connect each elementary particle to the singular boundary of its own unique inflating/deflating observable universe, with the divergent IR energy waves that appear to be emanating from the retarded manifestations of other elementary particles.

In this model it will be assumed that asymmetries in the density of IR potential energy in the retarded and null regions of spacetime will affect the trajectories of elementary particles in the observable universe in the following ways.

1) By the creation of asymmetries in an elementary particle's momentum space due to the scattering, absorption and re-emission, reflection of converging luminal or subluminal elementary particles or ensembles of elementary particles.
2) By the relativistic time dilation of signals that pass through or around more dense regions of retarded IR potential energy or by the relativistic time contraction of signals that pass through or around less dense regions of retarded IR potential energy which in the case of the former will increase the apparent distance between a particle and the singular boundary of the universe in the direction of the largest retarded IR potential energy density and in so doing make it appear that the universe older/colder/less dense/larger/more red shifted/more entropic in that direction and in the case of the later decrease the apparent distance between a particle and the singular boundary of the observable universe and in so doing make it appear that the universe is younger/hotter/more dense/smaller/less red shifted/less entropic in that direction.
3) With respect to the interaction of an elementary particle's filament with the converging and diverging super luminal IR energy waves of another elementary particle, virtual real and imaginary null forms of IR energy, which would alter the probabilities of where particles are likely to be observed in spacetime.

Examination of the possible IR energy symmetries and the effects they have on the evolution of the observable universe will be carried out with respect to the understanding that the observable universe emerged out of nothing via the breaking of perfect IR symmetry (the breaking of superposition) that gave rise to distinct observable regions of the IR universe that contain different flavors of IR potential energy.

In this naïve preliminary IR model of spacetime it will be assumed that the potential energy inherent in all elementary particles is a localized manifestation of the imaginary and real components of their entangled convergent and divergent IR energy waves. In this model the real component of IR potential energy will be used to model the behavior of what traditionally has been called electric charge and the imaginary component of IR potential energy will be used to model the behavior of what traditionally has been called gravitational mass.

It will also be assumed that these two forms of IR potential energy can be further bifurcated as follows.

i) Into a real positive component of IR potential energy, and a real negative component of IR potential energy (what used to be called positive and negative electric charge).

ii) Into an imaginary positive component of IR potential energy and an imaginary negative component of IR potential energy (what used to be called gravitational and anti-gravitational mass.)

This gives rise to the possibility of at least four manifestations of IR potential energy ($\pm E_{re},\pm E_{im}$), i.e., ($+E_{re},+E_{im}$), ($+E_{re},-E_{im}$), ($-E_{re},+E_{im}$), ($-E_{re},-E_{im}$), and to the possibility that these four types of IR potential energy gave rise to various inner and outer, anti-regions of spacetime after a sequence of primordial annihilation events broke the symmetry of the original IR null singularity i.e., disrupted the original superposition state of the universe.

Note that these negative and positive imaginary forms of potential energy are problematic with respect to traditional complex number theory because in traditional complex number theory complex numbers are represented as the sum of a real component and an imaginary component that somehow exist in a two dimensional complex plane defined by orthogonal real and imaginary coordinate axes, in which the complex coordinates of points in the complex plane are

somehow represented by the sum of the real and imaginary coordinates and not as ordered pairs i.e., in two dimensions as ($\pm x_{re} \pm x_{im}$). This strange representation of complex numbers as the sum of a real number and an imaginary number makes it impossible to order the imaginary component of complex numbers, i.e. to say that (1+i) > (1-i).

The bifurcation of the universe into two repelling inner and outer observable regions of the universe each of which contains one flavor of imaginary potential energy (mass) and two flavors of real potential energy (electric charge) allows the gravitational and electric forces of nature within those regions to be modeled using a single IR force law in the classical limit. Note that in the classical limit the effects of IR potential energy can be naively represented by an IR force law that is derived from the presence of IR potential energy, in which the imaginary component of the IR force always gives rise to an attractive force when the distance between like imaginary charges (mass) is observed to be real and a real component of IR potential energy (electric charge) can be attractive or repulsive when the distance between charges is observed to be real.

$$F_{IR} = k\frac{({q_{r1}}^{\pm})({q_{r2}}^{\pm})}{{R^2}_{elec}} + G\frac{({m_{i1}}^{+})({m_{i2}}^{+})}{{R^2}_{mass}} \quad or$$

$$F_{IR} = k\frac{({q_{r1}}^{\pm})({q_{r2}}^{\pm})}{{R^2}_{elec}} + G\frac{({m_{i1}}^{-})({m_{i2}}^{-})}{{R^2}_{mass}} \qquad [10]$$

Note that the square of imaginary same flavor mass will always produce a negative or attractive force in regions where the distance between the masses is expressed in real units and the possibility exists that the sign of these forces will flip in regions where the distance between the masses and charges is imaginary i.e., in imaginary inner spaces like gravitational masses will repel each other and like electric charges will attract each other; assuming of course that the imaginary masses do not become real masses and the real electric charges do not become

imaginary electric charges (mass becomes charge and charge becomes mass) when elementary particles transition from outer to inner regions or from inner to outer regions.

In this form of numerical analysis, traditional complex numbers are replaced by IR ordered n-tuples called IR coordinates and both real and imaginary coordinates can be assigned an order and anything goes. For example, an anti-IR space could be created by setting $(-1)^2 = -1$ and $i^2 = +1$. Further, the scale of the real and imaginary axes can be stretched (inflated) or compressed (deflated) in an infinite number of ways. Keeping in mind that the IR metric or distance function that is used to determine the distance between n-tuples in the IR space will have to be calibrated in order to construct precise mathematical models of the behavior of elementary particles and ensembles of elementary particles in the region of the universe we call "the observable universe."

An important concern here is related to the assumption that the physical properties of charge and mass are localized and to the commonly made erroneous assertion that the mass and electric charge of an elementary particle cannot be delocalized. The fact is that in order to experimentally confirm the locality of mass and electric charge one would have to measure two physical properties at the same time which is prohibited by the Heisenberg uncertainty principle. The fact is that if one measures the electric charge of an elementary particle and then measures its mass or vice versa, the first measurement will affect the position of the elementary particle which implies that the location of the second property will always appear to be in a different place in spacetime, which implies that the localization of the electric charge and mass of an elementary particle cannot be experimentally confirmed or refuted, which means that gravity could be a manifestation of electric charge, which would go a long way towards explaining why no one has been able to unify these two important fundamental forces of nature.

Relativity and Elementary Particles in an Infinite Potential Well.

In the traditional solution of a particle in a potential well in which the boundaries appear to be adiabatically moving away from or towards an elementary particle contained within it, little attention is given to relativistic concerns. If we ask the question why does an elementary particle contained in an asymmetrically expanding one-dimensional potential well appear to move towards the boundary that appears to be moving with the greatest velocity and away from the slower moving boundary relative to an observer outside the potential well? We come to the realization that for the case when the boundaries are moving with a velocity much less than the velocity of light that the motion of the observer relative to the boundaries of the potential well determines which boundary appears to be moving fastest and which appears to be moving slowest and to the realization that relative to an observer comoving with the elementary particle contained in an adiabatically asymmetrically expanding or contracting potential well the elementary particle will appear to move in such a way that it will re-establish the symmetry of the potential well by seeking to position itself in an observable universe in which both boundaries of the potential well appear to be receding or contracting with the same velocity. In the special case in which one boundary appears to be at rest and the other appears to be in motion with a velocity -v, such that -v is much smaller than the velocity of light, the final position of the elementary particle will reside at the center of a symmetric potential well defined by boundaries that appear to be receding with a velocity approximately equal to -1/2v relative to all observers co moving with that elementary particle.

If the observable universe is represented by a one-dimensional infinite potential well in which the boundaries are receding or contracting at relativistic velocities or at the velocity of light or at faster than light, virtual or imaginary velocities, then the theory of relativity demands

that we use the non-linear relativistic subtractive or additive velocity laws, which implies that the velocity of light will remain unchanged by the relative motion of observers contained in it. If the boundaries of the observable universe appear to be moving away from all observers at the velocity of light, then only the superluminal components of the convergent IR energy waves that emanate from the boundary of the observable universe will be able to manifest themselves as elementary particles, which correlates well with the predictions of traditional quantum theory in which the matter waves that give rise to elementary particles are considered to be imaginary and superluminal. Note that in this IR model, the superluminal IR energy waves contain both real and imaginary components which will impart elementary particles with the physical properties of mass and electric charge.

Thus, the effects of asymmetric distributions of retarded IR potential energy on the behavior of elementary particles can be summed up by the following quantum relativistic postulate.

> **Postulate 1**: An elementary particle at rest with respect to its symmetric boundaries will respond to the introduction of asymmetries in the rate at which its boundaries appear to be adiabatically expanding or contracting by accelerating towards the instantaneous symmetric center of its asymmetric observable universe and will continue to accelerate in that direction until the symmetry of its boundary conditions is re-established.

Note that the scales used to represent the real component of distance and the imaginary component of distance do not have to correspond in a one-to-one relationship, i.e., one imaginary meter could correspond to a thousand real light years. Which implies that the real and imaginary

components of converging IR energy waves do not have to be at the same energy level and for example, the system could be tuned to correspond to the observed differences in the magnitude of the forces they appear to impart to elementary particles.

The Particle in an Infinite Potential Well Revisited

The following simple thought experiment will be used to help visualize how asymmetrically arranged elementary particles in a potential well affect each other.

Let there be two identical elementary particles P_1 and P_2. Let elementary particle P_1 be trapped (ignore for now the possibility of tunneling) in an isolated symmetric one-dimensional potential well V_1 and elementary particle P_2 be trapped in an isolated symmetric one-dimensional potential well labeled V_2 (see figure 4).

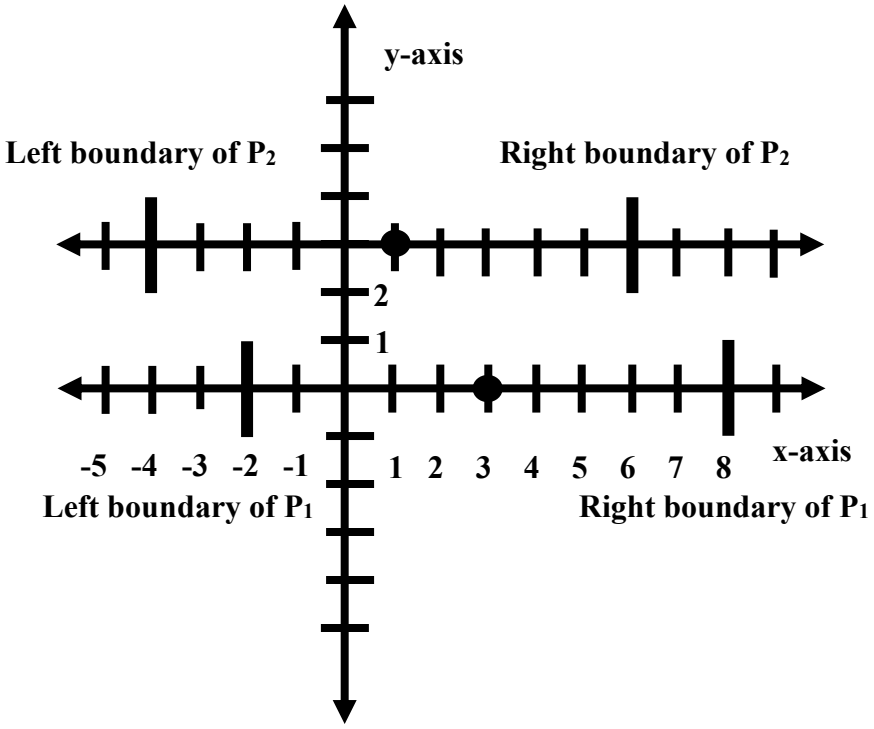


Figure 4

The most probable position of P_1 and P_2 before the two potential wells are transformed into a single hybrid potential well by adiabatically sliding the well containing P_2 into the well containing P_1 or vice versa in which case the particles will appear to move towards the point (2,3) as if acted upon by a mysterious or magical force.

Let each of the isolated symmetric one-dimensional potential wells be identical and bounded by a region of infinite potential. At this time, it will be assumed that the wave functions of elementary particle P_1 and elementary particle P_2 do not interact with each other, i.e., ignore for now the existence of exchange forces between them, and that elementary particle P_1 and elementary particle P_2 are in their respective ground states. Further, let these isolated potential wells exist in a two dimensional space represented by the (x, y) plane such that relative to an

observer O, the left and right boundaries of the potential well that contains P_1 are located at the points (-2,0), and (8,0) respectively and that the most probable ground state position of elementary particle P_1 in the potential well exists at the point (3,0); and relative to O, the potential well V_2 that contains P_2 is located on the horizontal line defined by the line (x,1) such that its boundaries are located at the points (-4,1) and (6,1) with the most probable ground state position of elementary particle P_2 is located at the point (1,1).

Now imagine what would appear to happen to the most probable position of particles P_1 and P_2 if the entire potential well V_2 is adiabatically slid into V_1 or vice versa (see figure 5).

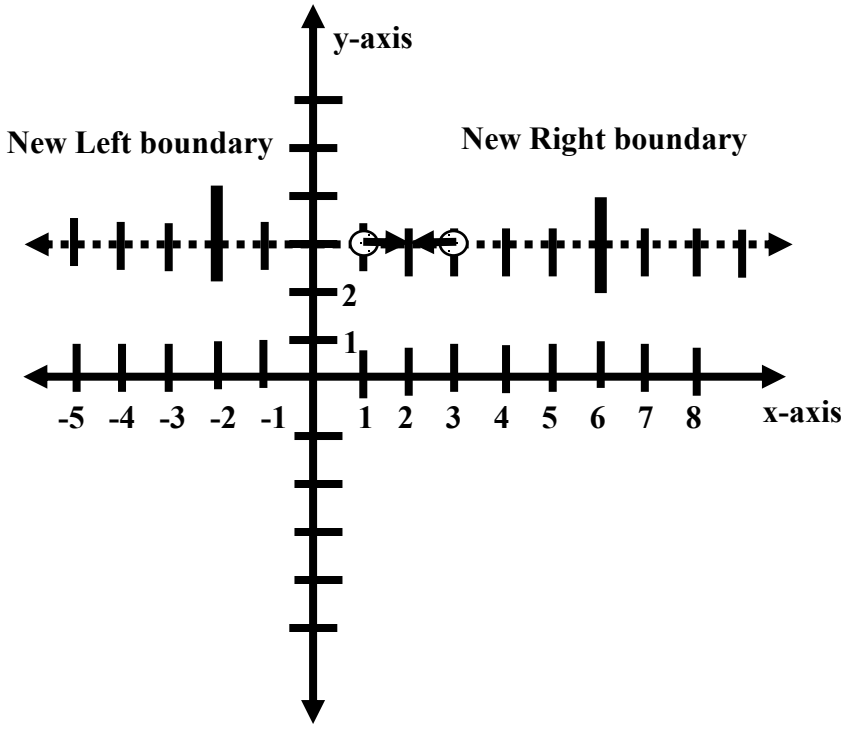


Figure 5

The direction P_1 and P_2 appear to move after the two potential wells are transformed into a single asymmetric hybrid potential well by adiabatically sliding the well containing P_1 into the well containing P_2, in which case the particles will appear to move towards the point (2,3) as if acted upon by a mysterious or magical force.

What the mathematics of quantum mechanics predicts is that relative to O, P_1 and P_2 would appear to slowly shift their position towards the center of the hybrid potential well V_3 formed by combining the boundaries of V_1 with the boundaries of V_2. Since the initial positions of P_1 and P_2 are no longer at the center of V_3, both P_1 and P_2 will appear to shift their positions until they arrive at the center of V_3 which in this case is located at the point (2, y), which in the

case of rigid boundaries implies that the particles will appear to move towards each other as the boundaries of V_1 and V_2 adiabatically merge to trap both particles between the left boundary of P_1 and the right boundary of P_2 which are located at the point (-1, 6) respectively, which will give rise to a smaller potential well, which in turn will increase the ground state energy of both the particles, and would make the particles appear to move towards each other as they each moved towards the center of the new potential well represented by V_3 with a motion that could easily be attributed to the presence of a magical or mysterious gravitational like force, or component of the gravitational force. Note that in this naïve model the boundaries V_1 and V_2 are treated as rigid walls but if they are not rigid, the possibility exists that both the left and right boundaries of the new potential well V_3 might adiabatically shift to a new hybrid left and right position that could maintain the original size of the potential well and thus keep the ground state energy of the two particles constant. Also note that

- the static boundaries of V_1 and V_2 could be replaced with boundaries that appear to be adiabatically receding from or moving towards the particles trapped between them or to be symmetrically or asymmetrically oscillating, as well as rotating.
- the observable universe relative to each observer in it is unique —Each observer resides at the vibrating symmetric center of a unique manifestation of IR spacetime and because of this all observed elementary particles are distinguishable, i.e., no two observers can observe the same particle they can only observe different manifestations of the same particle.
- any asymmetry in the distribution of IR potential energy will give rise to an expanding/deflating hybrid co-evolving potential well in which relative to each

observer in the hybrid potential well the rate of inflation will appear to be asymmetric.

- This type of model will have to be adapted to model the behavior of elementary particles in an observable universe in which all elementary particles are connected to their unique retarded non static (inflating/deflating) boundaries via convergent positive entropy IR energy waves but are not bounded by a future divergent boundary. An observable universe in which all elementary particles act as convergent IR energy wave sinks and divergent IR energy wave sources.

The Particle in a Potential Well Reimagined.

Examining the two-dimensional (ψ,x) potential well graph that traditionally has been used to represent the probabilities of locating particles in potential wells or harmonic oscillators (see figure 1), it is obvious that they do not include a fictitious dimension of time in the graphical representation; which precludes the possibility of unifying such depictions with classical and modern relativistic graphical representations of spacetime. They also fail to take into account that the IR energy waves that originate on the boundary of spacetime and give rise to what appears to observers in the observable universe to be elementary particles must themselves be quantized and must converge from the origin of spacetime and then diverge towards the future state of the observable universe; which implies that relative to a particular elementary particle all other elementary particles in the observable universe will appear to behave as if they are a concentrated source of quantized diverging IR energy waves and that the intensity of the quantized boundary IR energy waves diverging from a retarded source will appear be inversely proportional to the square of the luminal distance separating elementary particles in the observable universe. Finally, they fail to take into account the fact that elementary particles

possess an intrinsic quantized amount of spin which implies that the region of the IR boundary that defines the existence of a particular elementary particle in it must possess an intrinsic amount of angular momentum and the converging IR energy waves that give rise to the existence of elementary particles must contain quantized amounts of IR angular momentum.

Since at this time how the quanta that make up IR energy waves go from being convergent to being divergent is unknown i.e., are they reflected/scattered off the elementary particle or are they transmitted through elementary particles or some combination of the two; the models that are created using any of these underlying assumptions must all be structured in such a way that they highly correspond to the predictions of classical and relativistic theory when applied to the mathematical description of the macroscopic observable universe. For the sake of simplicity, it will be naively assumed that the IR quanta that make up IR energy waves are either reflected or scattered off of elementary particles such that the angle of incidence is equal to the angle of reflection or are equally transmitted through and reflected or scattered off the elementary particles they give rise to. Note, convergent and divergent IR energy waves have both real and imaginary components and that the divergent IR energy waves emanating from a retarded source will be less dense than the convergent IR energy waves that make up isolated elementary particles and that the combination of convergent IR energy waves and the divergent IR energy waves that appear to emanate from retarded asymmetrically distributed elementary particles will give rise to asymmetries in the density of the convergent IR energy waves that give rise to elementary particles and that this asymmetry in the density of the convergent IR energy waves will affect the group velocity and most probable position of an elementary particle relative to the retarded elementary particles that appear to be asymmetrically distributed around it. It is important to note that in a three-dimensional universe the quantized spherical convergent and

divergent IR energy waves will microscopically give rise to an infinite number of possible instantaneous PM vectors and that an elementary particle will appear to be macroscopically static but microscopically oscillating (vibrating) when the retarded arrangement of IR potential energy is symmetrically distributed around it and appear to be macroscopically moving in a preferred direction and microscopically vibrating when the retarded arrangements of IR potential energy is asymmetrically distributed around it.

In classical and modern relativistic theory, time is considered to be an actual physical dimension and in modern relativistic theory spacetime it treated like a deformable fabric, much like the jumping mat of a trampoline. As outlined earlier, the modern relativistic depictions of spacetime events utilizes null geodesics to causally connect the observable past states of the universe with future states, which implies that relative to an observer located at P_1 in figure 2, the present position of all the other elementary particles in the observable universe reside in a virtual inner space bounded by the horizontal light cone defining null geodesic boundaries that extend from the retarded origin of spacetime to the nexus of P_1 and from the nexus of P_1 to the future state of all the other elementary particles in the universe. Note, that with respect to figure 2 the filament of P_2 relative P_1, is represented by the line segment that extends from P_2^* to P_2^{**} and that relative to an observer located at P_1 the segment of the filament that extends from P_2^* to P_2 is completely determined i.e., has already existed in an unobservable region of the past and the line segment that extends from P_2 to P_2^{**} and represents all possible future outcomes of P_2 that cannot be altered by an observer at P_1 as it transitions from P_2 to P_2^{**}. How much of another elementary particle's existence is determined and unobservable or virtual relative to an observer at P_1 depends on how far in the past an elementary particle appears to exists relative to that observer. Also note that relative to an observer at P_1, P_2 exists in a virtual state of quantum superposition.

Although classical and modern relativistic fictitious space-time figures like figure 2 are useful they are of little use in traditional quantum theory where in the microcosm the probability that events will occur are depicted in terms of the interaction of super luminal matter waves and virtual fields of particles, and all possible outcomes are considered to be in quantum superposition until a measurement is actually made and it is for this reason that the field of physics bifurcated in a way that precludes the possibility of unification. What follows is an attempt to unify the field by using effective IR models refined from the one previously outlined with effective here implying that the theory ignores the obvious infinities inherent in an observable universe in which elementary particles are manifestations of converging quantized IR energy waves.

Let there exists two particles P_1 and P_2 that appear to be at rest with respect to each other relative to observers located at P_1 and P_2 in figure 3. It will be assumed that there is no observable motion with respect to any of the fictious coordinate axes used by P_1 and P_2 to model the spacetime continuum and that unlike modern relativistic theory, in this IR model of space the principle of relativity will be extended to include the dimension of time, which implies that the depiction of motion with or through a physical dimension of time or spacetime requires a fictitious coordinate system. Note that this must be true even if there actually existed a physical dimension of time that all things in the universe appeared to move along relative to a fictitious observer located at the origin of spacetime.

Removing the dimension of time leads to the realization that all comoving observers will appear to be nearly at rest with respect to each other (there will always be a whole lot of particle jitter and interaction to consider) and in terms of the trace IR coordinate system previously outlined the instantaneous unobservable trace IR distance separating P_1 and P_2 will be defined to

be a virtual (null) imaginary valued trace geodesic, and all other trace IR geodesics connecting P_1 and P_2 will contain both real and imaginary vector components (see figure 6).

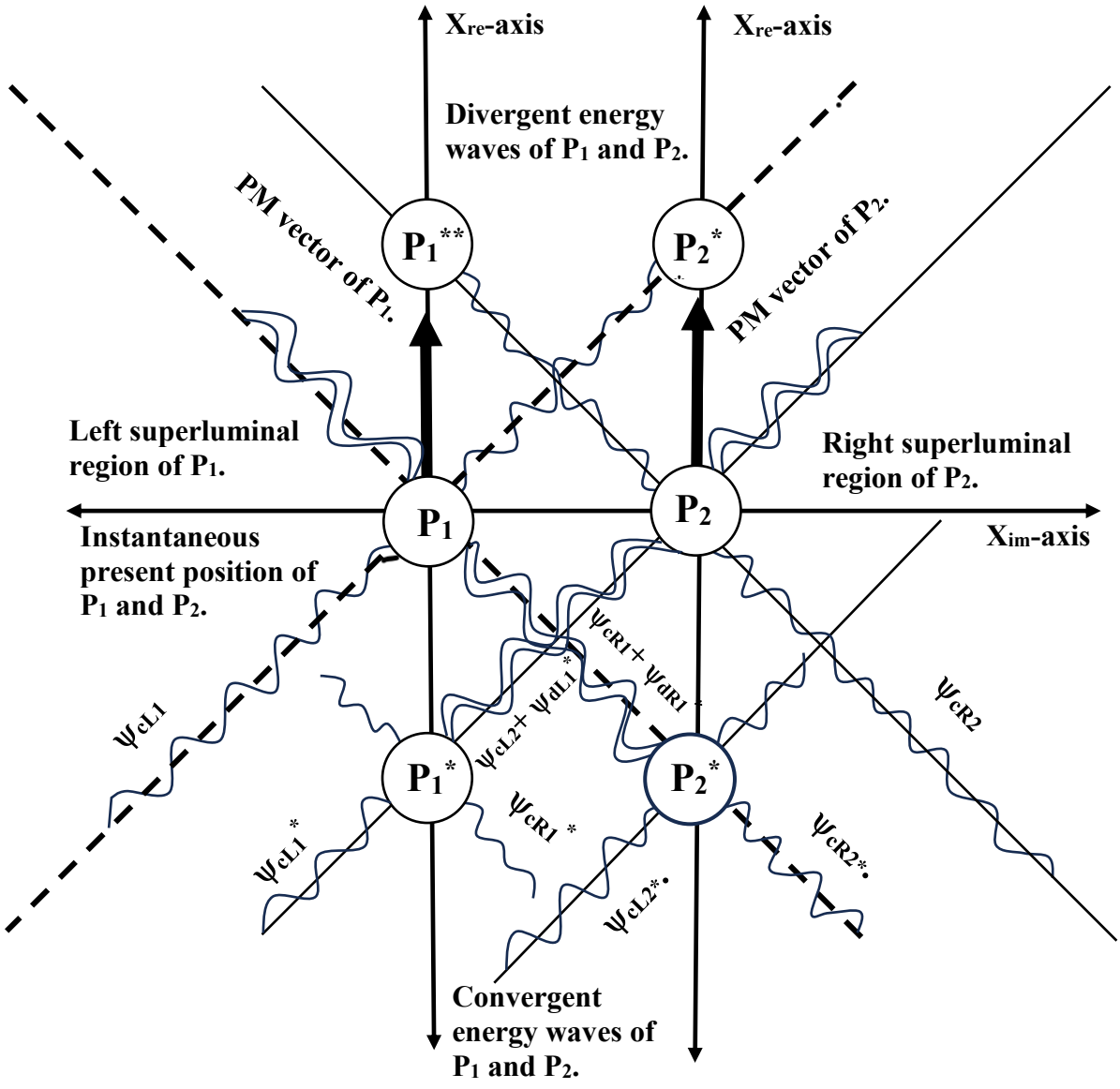


Figure 6

This Fictitious IR frame of reference represents the position of two co-moving particles P_1 and P_2 in the IR spatial trace plane. Note the interference of convergent and divergent IR energy waves can induce asymmetries in the most probable location of P_1 relative to P_2 and can even make it appear that the particles are moving towards or away from each other, depending on how the convergent IR energy waves of one particle interact with the divergent IR energy waves of the other particle. Note, the interaction of the IR boundary waves of P_1 and P_2 will cause the PM axes of P_1 and P_2 to rotate relative to each other.

Thus, all positive entropy super luminal actions on P_1 can be mapped to the imaginary inner region of the IR trace plane defined by the horizontal slice of a dual light cone with a nexus located at P_1.

In figure 6 let the trace luminal boundaries of space that separate real spatial intervals from imaginary spatial intervals relative to an observer at P_1 be represented by four null geodesics; two projecting upwards from P_1 at an angle of 45° one of which that connects P_1 to a future IR manifestation of P_2 labeled P_2^{**} and the other that points to future events in the opposite direction and another pair of geodesics that point downward from P_1 at an angle of 45°, one towards a retarded IR manifestation of P_2 labeled P_2^{*} and the other in the opposite direction. Note, that the angle of these geodesics has been chosen arbitrarily and in general they are dependent upon the metric used to calculate the distances between events of which 45° is just one of an infinite number of possible choices.

Thus,

- all geodesics in the IR trace plane have clocks associated with them.
- all events in the imaginary inner regions will appear to observers in outer regions to occur in null intervals of time i.e., will appear to outer observers to occur simultaneously, which implies that relative to observers in outer regions, the combined effect of all the boundary IR energy waves that pass through these inner imaginary IR regions can be summed into a single IR vector.

The construction of models that can relate what is observed to be taking place in the outer observable regions of the universe in terms of processes that are taking place in the inner imaginary virtual regions of space will require that what we imagine to be taking place in the virtual regions correspond to what we observe to be taking place in the outer real regions of

space, while simultaneously keeping in mind that what we think we know about the physical universe is inextricably linked to the evolution of technology and the limits placed on the precision of measurements by the Heisenberg restrictions of quantum mechanics. Thus, the successful development of an evolving unified field theory depends upon the scientific community being able to clearly answer two very important questions:

i) What do we know?

ii) To what degree of precision can we claim to know it?

The construction of the following model of spacetime will be based upon the following well established facts about the observable universe:

1) Electric charge appears to come in two flavors in the observable universe.
2) There are equal numbers of positive and negative electric charge in the observable universe (charge parity).
3) Luminal bosons have no rest mass.
4) Luminal bosons are indifferent to the presence of electric charges but are transversely deflected by the presence of matter by twice the amount predicted by classical gravitational theory.
5) The deflection of luminal bosons by matter is correctly predicted by the General Theory of Relativity but the General theory of relativity was developed prior to the development of quantum theory and the discovery of anti-matter.
6) The electric and gravitational influences of particles that are observed to be at their retarded positions, appear to originate from what would be the imaginary contemporary position of that particle, as if spacetime somehow knows where they actually are in the unobservable inner region of spacetime.

7) Even in indeterminate probabilistic quantum theories, the probability as to where an elementary particle might be observed in spacetime is temporally restricted such that elementary particles do not have observable probabilities of appearing in the past. The reason for this restriction is related to the understanding that the universe contains equal numbers of positive and negative electric charges (charge parity). If it were possible to build a device to send elementary particles back into the past or if particles could randomly appear in the past, then in the former case it would be possible for a scientist to send a large number of electrons back in time or in the latter case at any time a single electron might return to the past which in either case would change the charge balance of the both the present and past universe. If such a thing were possible a person could use a series of electrons and positrons to send morse code signals back to themselves in the past.
8) Particle/wave duality. What appears to us to be particles or waves are neither, which implies that the particle description and the wave description of nature are fabrications of the human mind and although useful, neither without the other is capable of producing a coherent model of the observable universe. Note that quantum theories in which elementary particles are treated as being entangled manifestations of matter waves are mute as to what to call the quanta that make up matter waves.
9) Because each elementary particle in the universe exists in a unique subset of the universe called an observable universe, the manifestation of each particle relative to each unique observable universe are themselves unique, which implies that there are no such things as indistinguishable or identical particles, or perhaps that

all the particles of the same type in the observable universe are just unique manifestations of the same singular field boundary, i.e., there is only one electron and an infinite number of unique ways to observe it.

At first glance these nine facts might seem unrelated to each other, but to flesh out the connection between them will require us to hitch a ride on a photon.

In traditional relativistic theory it is impossible to transform a photon into a rest frame, and such transformations are labeled to be forbidden transformation. However, this transformation can be accomplished using quantum theory in which a photon can be transformed into a time static but orientated annihilation event i.e., into a timeless superposition of an outer particle/inner anti particle pair in which the momentum of the outer particle points in the direction the photon was observed to be moving relative to an observer whose frame of reference defines the limits of the observable outer region and the other photon's momentum vector is virtual, which implies that only the photon that has its momentum vector pointing in the direction the original photon was moving can interact with the IR potential energy that exists in the outer region of space the original photon was observed to be moving through, while the other photon exists as a virtual photon that will not interact with matter in the outer observable universe.

In addition, the elementary particles in the observable universe that photons appear to be moving at the speed of light relative to will need to be transformed into photons that contain zero rest mass and the only way to do this is to combine them with their antiparticles. What this implies is that relative to a photon, the universe in the direction of its motion appears to be a time static annihilation event and the universe in the opposite direction is transformed into an unobservable inner region. This implies that relative to the energy contained in a photon the universe will appear to contain zero electric charge, and twice the amount of energy (mass) that

it appeared to contain relative to an observer moving at sub luminal speeds. Since we know that the magnitude of a force producing field of IR potential energy acting on a photon in the direction of its motion is unchanged from its classical value and the distance between points in the direction of its motion is zero, only the transverse components of the superposition (annihilation) event will be able to affect the geodesic of the original photon. This implies that there exists a superposition symmetry between what we call the observable universe and its anti-universe and that the velocity at which a particle moves can affect the virtual distance separating these two distinct manifestations of IR potential energy (mass and charge) in the IR observable universe. This also implies the possible existence of an axis of symmetry between what we call gravitational potential energy (the imaginary component of potential energy) and a type of anti-gravitational potential energy (an anti-imaginary component of IR potential energy).

Using the imaginary component of IR potential energy to represent what is presently called the rest mass of elementary particles and the real component of IR potential energy to represent what presently is called the electric charge of elementary particles gives rise to the possibility of IR potential energy and anti-potential energy (matter-antimatter) and partially symmetric IR forms of IR potential energy i.e., energy/anti-energy symmetries of the form $(E_{re}{}^{\pm}, E_{im}{}^{\pm})$ with the ($E_{im}{}^{+}$) type of imaginary potential energy repelling the ($E_{im}{}^{-}$) type of imaginary potential energy, but particles with components of imaginary energy with like sign attracting each other. Note this implies the existence of at least two distinct observable regions of the universe that would have gravitationally repelled each other early in the history of spacetime and that exist as outer/inner regions of spacetime relative to each other and that the repulsive interaction between these two observable inner/outer universes might be one of the main driving forces behind the inflation/deflation of the observable universe. Additionally, the

breaking of this symmetry (superposition) may have preceded the breaking of the symmetry (superposition) between the real positive component of IR potential energy (E_{re}^{+}) which presently is called positive electric charge and the real negative component of IR potential energy (E_{re}^{-}) which presently is called negative electric charge.

It was found that it is possible to simplify the graphical presentation of the net interaction between two comoving particles created by three dimensional converging and diverging IR energy waves in regions of the observable universe that arose from the breaking of these more inclusive IR symmetries, by utilizing a two dimensional IR probability trace plane in which the probability of finding a particle P_1 and the orientation of its fictitious PM vector in the trace plane is inferred and calculated from the interaction of the left and the right components of its converging IR energy wave (ψ_{cL1}) and (ψ_{cR1}) respectively in figure 6, with the left diverging (ψ_{dL2}) IR energy waves of another particle P_2 located to the right of P_1 in figure 6.

In the absence of other particles or in the case of a symmetrical arrangement of IR potential energy in the retarded observable universe, the left and right IR trace probability components of a particle's PM vector are equal and the particle will, over macroscopic intervals of time, appear to be nearly at rest relative to other comoving particles, i.e., elementary particles that share the same degree of retarded IR potential energy symmetry. Note that even though a particle may over macroscopic intervals of time appear to be at rest, on the microscopic level that particle may appear to be oscillating, and that this oscillation may also include a virtual component.

When an IR potential energy asymmetry exists in the retarded spacetime that surrounds a particle, the left and right components of a particle's IR energy wave will no longer be equal which will alter the probability of where that particle is most likely to be found in the IR trace

plane relative to other particles in the plane, which could be represented by a rotation (a boost) of a particle's fictitious PM vector relative to the PM vector of other particles in the observable universe.

In order to utilize this type model to explain the effects of the presence of what presently are called electric forces on the probability of locating particles in the IR trace plane, one needs to be able to explain why like charges appear to repel one another and opposite charges appear to attract one another. It was found that this can easily be accomplished by assigning the positive real component of an elementary particle's IR energy wave a positive phase ($\psi_{Re}{}^{+}$) and the negative real component of its IR energy wave a negative phase ($\psi_{Re}{}^{-}$) and allowing the phase of the real components of the IR energy waves of different elementary particles to be constructively and destructively added together. Note, the IR energy waves in both the real regions and imaginary regions of IR space have both real and imaginary components and some of those components will transfer kinetic energy via observable collisions with elementary particles and when asymmetric arrangements of IR potential energy are present in the retarded regions of IR space relative to a particular elementary particle, the asymmetric arrangements of IR potential energy will give rise to a ubiquitous Le Sagian shadow push force.

In the case of what presently are called repulsive electric forces, let the instantaneous virtual present of two manifestations of the positive real component of IR potential energy (positive electric charges) $q_1{}^{+}$ and $q_2{}^{+}$ be represented by the imaginary line segment connecting $q_1{}^{+}$ and $q_2{}^{+}$ (see figure 7).

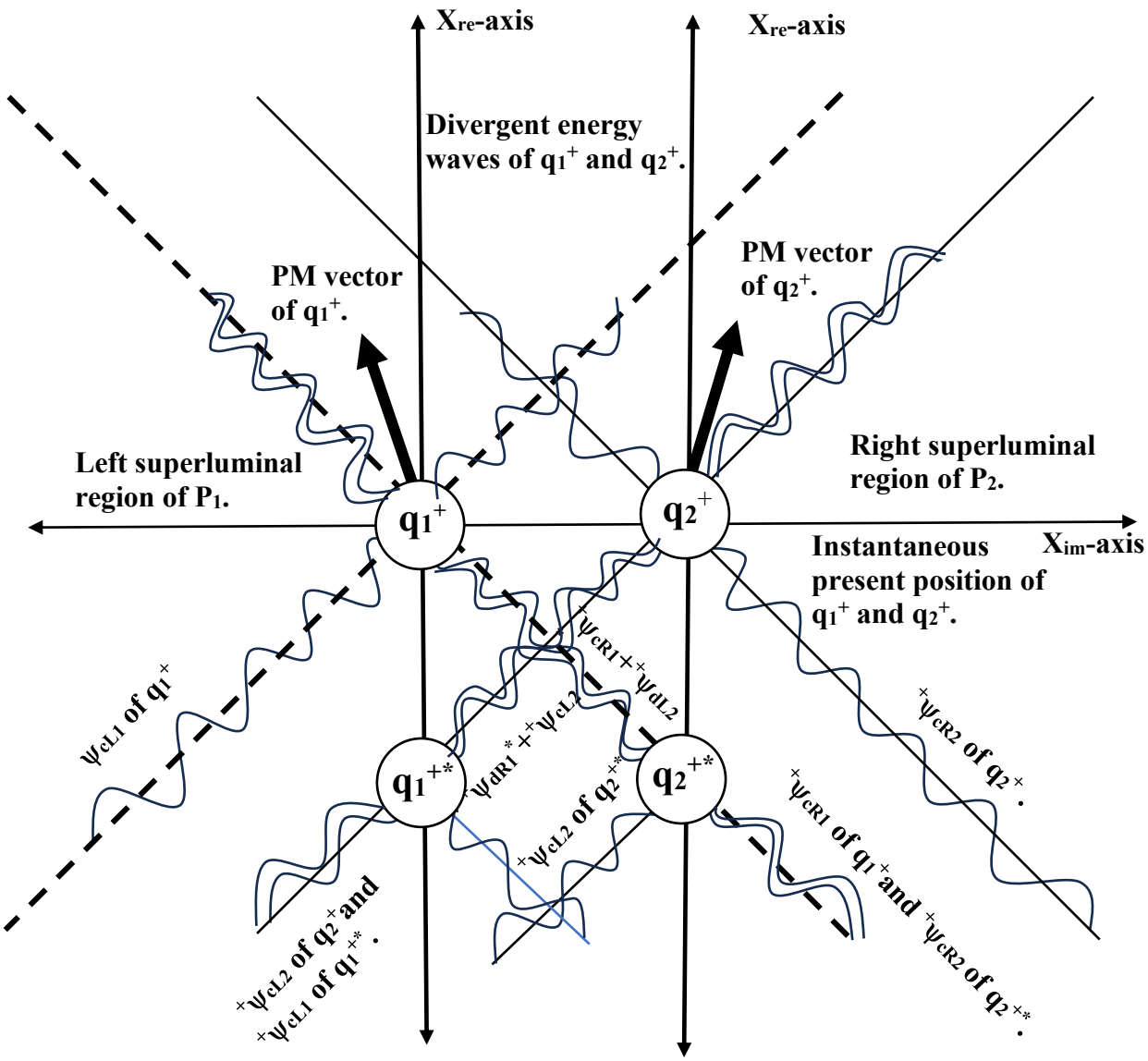


Figure 7

Fictitious IR frame of reference representing the contemporary IR spatial trace position of q_1^+ and q_2^+ and how the convergent IR energy waves of the charged elementary particles interact with the divergent IR energy waves emanating from the retarded manifestations of other electric charges to make it seem that they are being motivated to move away from each other by a mysterious force.

Let the retarded luminal null position of q_2^+ relative to q_1^+ be given by q_2^{+}*. Assuming the phases of the real components of IR energy waves are additive then it is important to note that:

1) In an inflating observable universe the retarded boundary of q_2^{+}* represents a region of the boundary of the universe that existed at a time when the universe was smaller and therefore higher in energy than it is relative to the present position of q_1^+.

2) Although the luminal boundary of q_2^{+}* appears to intersect the luminal boundary of q_1^+ this is just a graphical illusion i.e., the boundaries of each elementary particle do not coincide at the same time.

3) Although the luminal boundary of q_2^{+}* appears to be symmetric in the figure it could in fact be, and in all probability is asymmetric i.e., being influenced by the diverging IR energy wave scattered by a retarded manifestation of q_1^{+*}.

4) Although the right luminal boundary of q_1^+ appears to coincide with the right luminal boundary of q_2^{+}* all the way back to the beginning of the observable universe these geodesics do not exist at the same time.

5) The left divergent component of the IR energy wave diverging from q_2^{+}* represented by $^{+}\psi_{dL2}{}^{*}$ in figure 7 will interact with the right component of the convergent IR energy waves ($^{+}\psi_{cR1}$) to create an asymmetric IR energy wave of q_1^+ such that the energy of $^{+}\psi_{cR1}$ is raised to ($^{+}\psi_{cR1} + {}^{+}\psi_{dL2}{}^{*}$) which is greater in energy than $^{+}\psi_{cL1}$ which will make it seem like q_2^+ is repelling q_1^+ and vice versa for the effect that the divergent IR energy waves emanating from q_1^{+*} have on the current position of q_2. Note that this will also be true if q_1 and q_2 are both negatively charged. When q_1 and q_2 are oppositely charged, for example when q_1 is positively charged and q_2 is negatively charged, the phase of the IR energy wave converging from the right

towards q_1^+ and the phase of the IR energy wave diverging from q_2^-* will cancel each other out, which will create an asymmetry in the energy density of the IR energy wave converging towards q_1 such that ${}^{+}\psi_{cL1} > ({}^{+}\psi_{cR1} + {}^{-}\psi_{2dL}{}^{*})$, which will make it appear that some kind of magical force is pushing q_1^+ and q_2^- together.

Although this way of depicting the interactions that give rise to what traditionally has been called the Coulomb force acting between a static source charge and another electric charge is promising much is missing. Three very important things that are missing are;

1. The ability to handle the general case when two charges appear to be in motion relative to an observer.
2. The inability of the model to account for the fact that relative to luminal bosons the transverse component of the gravitational force will appear to be twice the value that it appears to be for objects that are moving much slower than the velocity of light, while the electric force acting on luminal particles (photons) is zero, i.e., the geodesics of photons are not affected by the presence of electric charge.
3. The need to take into consideration the effect that quantized spin has on the interaction of elementary particles.

Although as previously noted it should not be difficult to adapt this type of model to include the hyperbolic spacetime distortion induced by relative motion as already derived in the special theory of relativity, much more will have to be taken into consideration in order to develop a model that can handle the implication of 2 and 3 in the above list of things that need to be taken into consideration.

With respect to 2 above it will need to be assumed that relative to photons all the elementary particles that appear to be charged in the sub luminal observable universe will appear in the luminal frame of reference to be in superposition with their anti-particles i.e., appear to be a charge symmetric annihilation events, and since relative to photons the universe is singular along its direction of its momentum, real energy symmetry is re-established when the distance between points in the direction of a photon's momentum is zero and the time between emission and absorption is zero. To represent this superposition, the real component of a particle's IR potential energy will be represented by a pair of particle-antiparticle pairs in the following way (see figure 8).

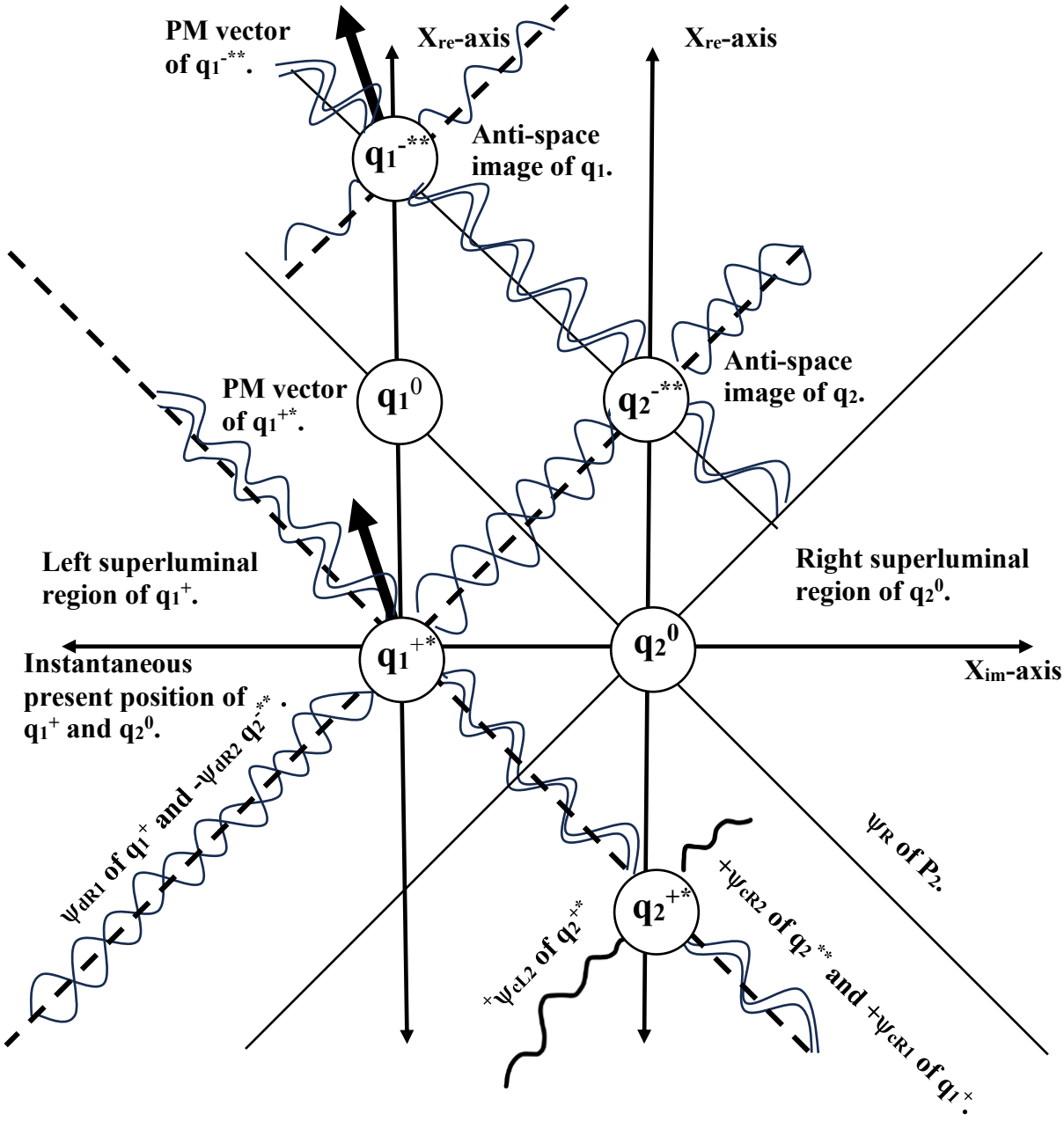


Figure 8

Fictitious IR frame of reference representing the IR trace position of q_1^{+} relative to q_2^{0} which represents the inner or virtual superposition of (q_2^{+*} and q_2^{-**}), which represents a repulsive arrangement.

In figure 8 the retarded position of q_2^{*} relative to q_1^{*} and q_2^{**} relative to q_1^{**} is represented by the causal luminal connection between q_1^{+}* and q_2^{+}* and the causal luminal anti-connection between q_1^{-}** and q_2^{-}**, while the causal luminal connection between the superposition states q_1^{0} relative to q_2^{0}, represents the interaction between a superimposed pair of charge symmetric particles that contains zero real potential energy (electric charge) and twice the rest imaginary potential energy (mass) of the particles that are in superposition. Since in this case q_1^{*} and q_2* represent positive manifestations of real IR potential energy and q_1** and q_2** represents negative manifestations real IR potential energy then it is expected that the phase of the real components of the IR energy waves will add along the line of luminal causality connecting q_1^{+*} and q_2^{+}* and along the line of luminal causality connecting q_1^{-**} and q_2^{-}** which implies that the sum of the PM vectors associated with q_1^{0} relative to the retarded position of q_2^{0} will point away from q_2^{0} and the sum of the PM vectors associated with q_2^{0} relative to the retarded position of q_1^{0} will point away from q_1^{0} — Which implies that they have a decreased probability of being observed to be moving towards each other and an increased probability of being observed moving away from each other and that they will appear to move towards each other when q_1^{*} and q_2* and q_1^{**} and q_2** represent opposite manifestations of real potential energy (see figure 9).

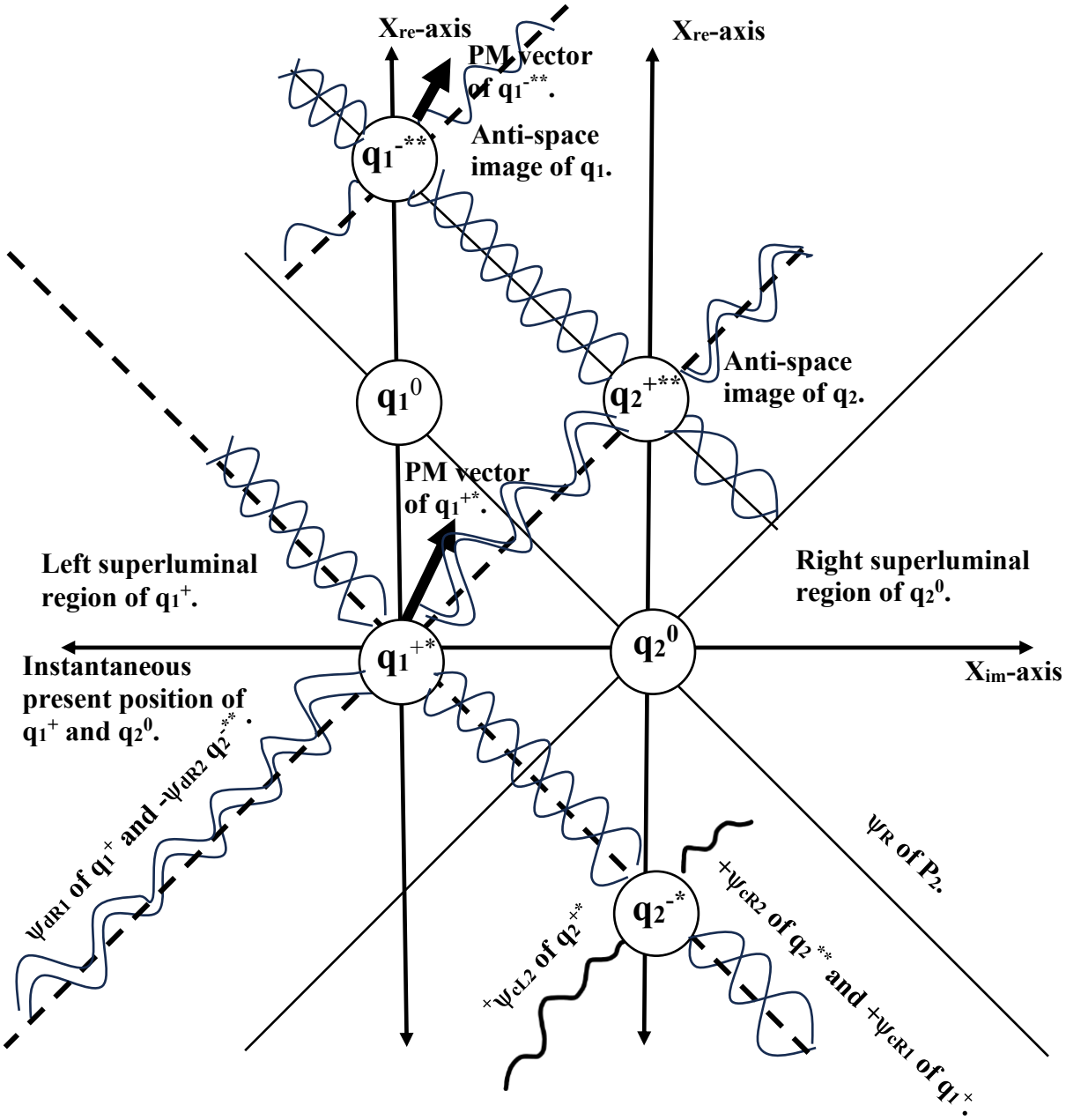


Figure 9

Fictitious IR frame of reference representing the IR trace position of q_1^{+*} relative to q_2^{0} which represents the inner or virtual superposition of (q_2^{-*} and q_2^{+**}), which represents an attractive arrangement.

The amazing thing about this conceptualization of space is that it can connect relativistic and quantum theory by introducing the idea of anti-space/anti-curvature or anti-imaginary potential energy and the idea that space and anti-space can exists in superposition relative to time static geodesics, into the General Theory of Relativity and thus can be used to predict the correct motions of elementary particles and ensembles of elementary particles using a variety of relativistic quantum mechanical models i.e.,

- by assigning phase relationships to the IR potential energy waves that give rise to elementary particles in the observable universe.
- by allowing asymmetries in the retarded arrangements of elementary particles about an elementary particle to alter the inflation/contraction rate of spacetime; with elementary particles appearing to move towards the longest/oldest/most dense/coldest/most entropic/most red shifted direction of the observable universe and away from the shortest/youngest/least dense/hottest/least entropic/least red shifted direction of the observable universe,
- by assuming that asymmetries in an elementary particle's convergent IR energy wave that gives rise to elementary particles can affect the tenacity with which an elementary particle couples to the Higgs field, or the density of the Higgs field, with elementary particles always appearing to move in the direction of the weakest coupling to the Higgs field.
- by allowing IR forms of potential energy to be annihilated by anti-IR forms of potential energy or partially annihilated by forms of IR potential energy that contain the same type of either real or imaginary potential energy, and treating the

orientation of an elementary particle's fictious PM vector in the IR trace plane as a vector sum of its left and right components of its convergent IR energy wave.

This leads to the following very important questions.

Are all these possible ways of describing the motion of elementary particles in the observable universe distinct from one another or are they just different ways of saying the same thing? And if they are, is there a general theoretical framework that can reduce them to a single unifying explanation?

At this time the explanation that seems to be most promising is that asymmetries in a contemporary elementary particle's convergent IR energy wave will alter the most probable position and group velocity of that contemporary particle by making it appear that the contemporary particle does not reside at the center of its observable universe i.e., that the rate at which the contemporary particle's boundary appears to be inflating or contracting is no longer symmetric, which implies that the description of the effects that asymmetric retarded manifestations of IR potential energy have on the most probable position and group velocity of a contemporary elementary particle in the observable universe can be reduced to calculating the effect that divergent (inflating) IR energy bubbles emanating from asymmetrically distributed retarded manifestations of IR potential energy have on the symmetric convergent IR energy waves of a contemporary elementary particle.

Conclusion:

Although this IR model of spacetime and the use of inflating and deflating regions of pixilated IR spacetime shows promise in unifying the explanation of the naturally occurring forces of nature much still needs to be considered. High on the list of things that will need to be considered are:

- The concept of spin/isospin and the concept of exchange forces that are known to arise between identical elementary particles, i.e., fermions and bosons.
- If there is a unifying relationship between the electric force acting between negative or inverted inner space confined quarks and what is presently is called the strong force (electro-strong unification).
- If what is presently called the gravitational force is a residual electric and dipole-dipole force that arises because of quantum exclusion, the Heisenberg uncertainty principle, asymmetries in the most probable distribution of the electrons and quarks in one atom relative to the electrons and quarks in other atoms, and the differences between the mass of electrons and protons —And if the gravitational force is a manifestation of the electric force then does the need for the separation of electric charges within a black hole imply that true singularities cannot exist?
- How IR energy waves and the boundary conditions of the observable universe give rise to elementary particles with quantized spin states and at what degree of precision will the Heisenberg uncertainty relationship between angular momentum/rotational energy and position reduce the predictive power of quantum theory to the point that the underlying chaotic nature of spacetime becomes transparent.
- How to quantize the IR energy waves and what to call the unobservable elementary quanta that make up the IR energy waves that give rise to what appears to us to be elementary particles.

Although I think that this work reveals that it is possible to explain the naturally occurring forces of nature and the pixilated nature of spacetime in terms of the incessant asymmetric pinging of quantized converging and diverging IR energy waves that connect

elementary particles with the singular origin of spacetime and also increases the number of mathematical tools that can be used to describe the observable IR universe, as well as many other possible observable universes, much mathematical work will need to be done to develop coherent, non-Argandian forms of IR statistical analysis that can be used to model the behavior of all the manifestations of IR energy/entropy in the observable physical universe and the universe out of which it evolved.

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Fictitious IR frame of reference representing the contemporary IR spatial trace position of q_1^+ and q_2^+ and how the convergent IR energy waves of the charged elementary particles interact with the divergent IR energy waves emanating from the retarded manifestations of other electric charges to make it seem that they are being motivated to move away from each other by a mysterious force.

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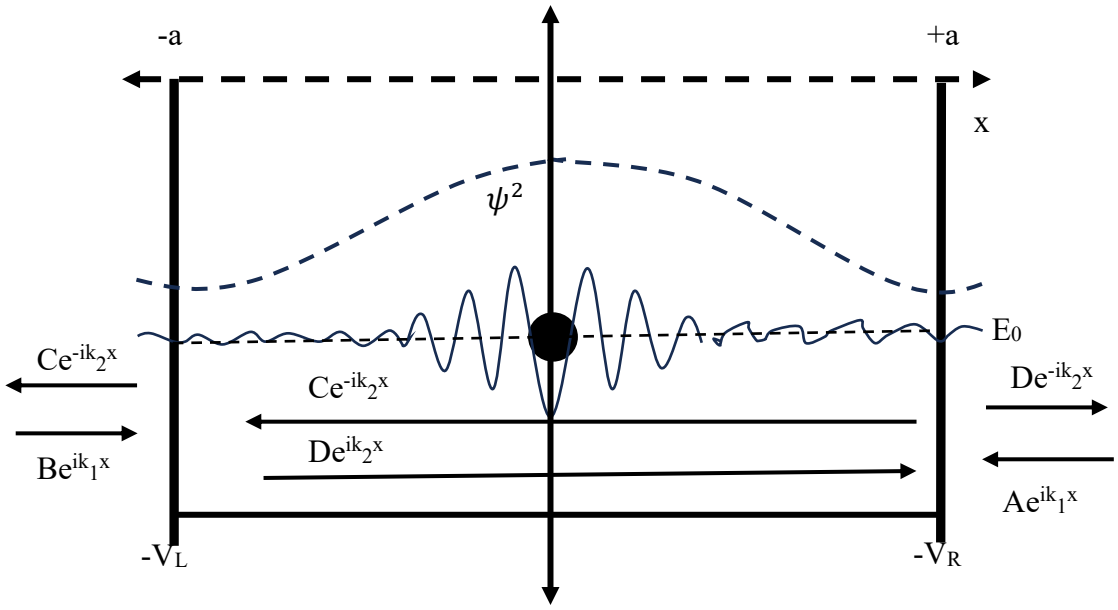


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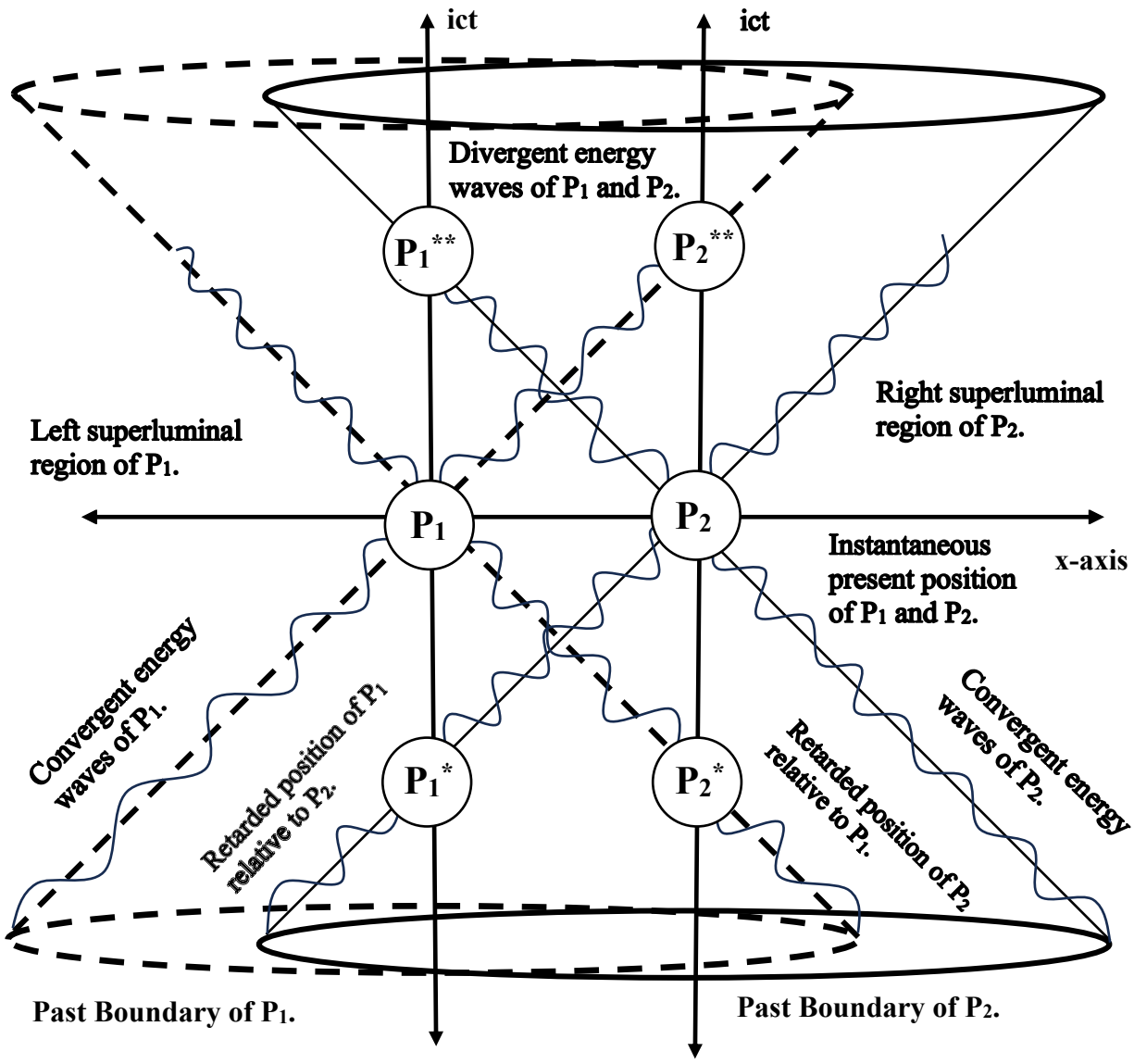


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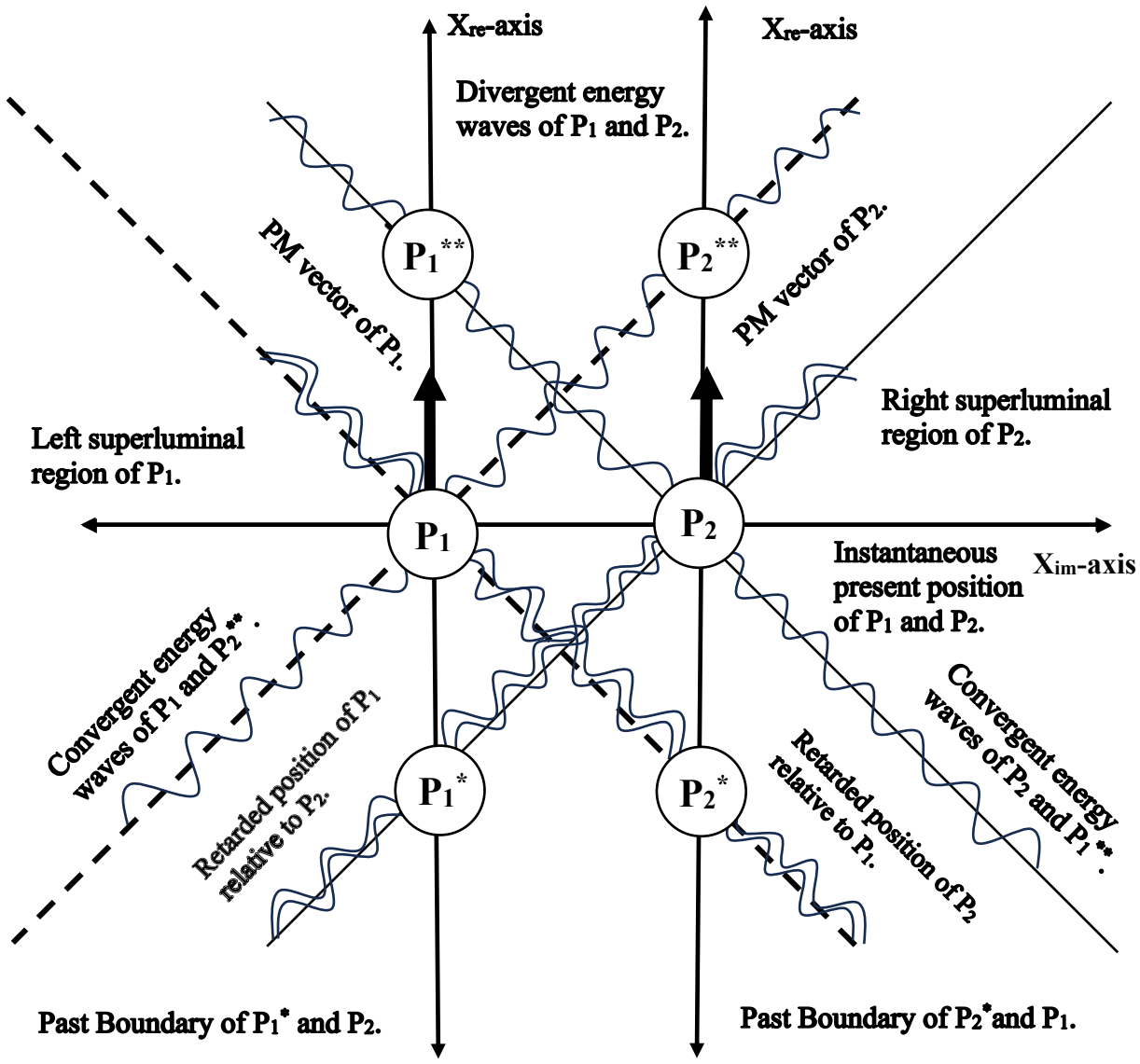


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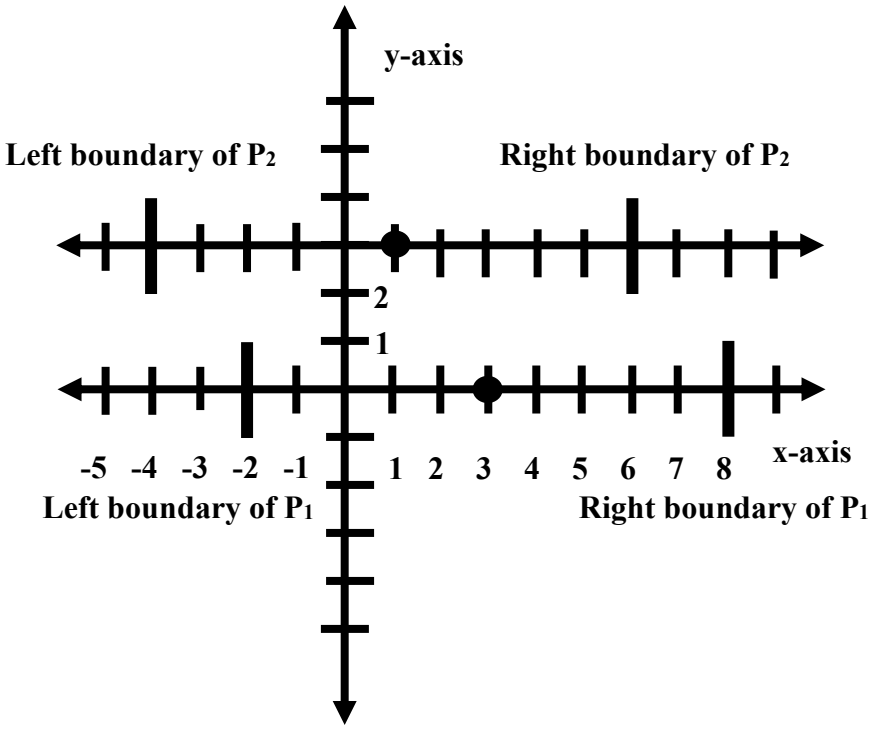


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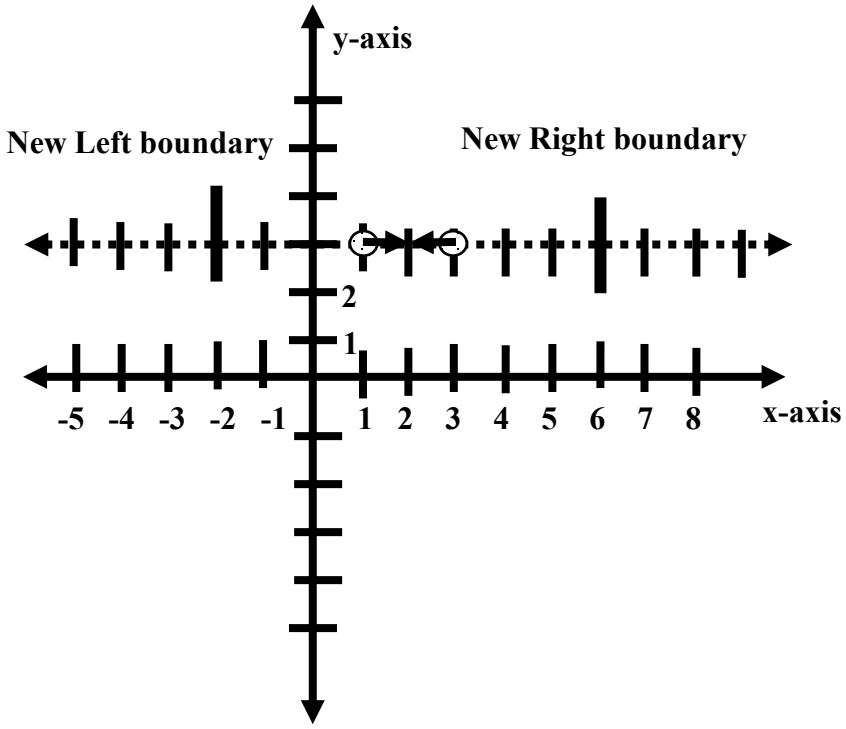


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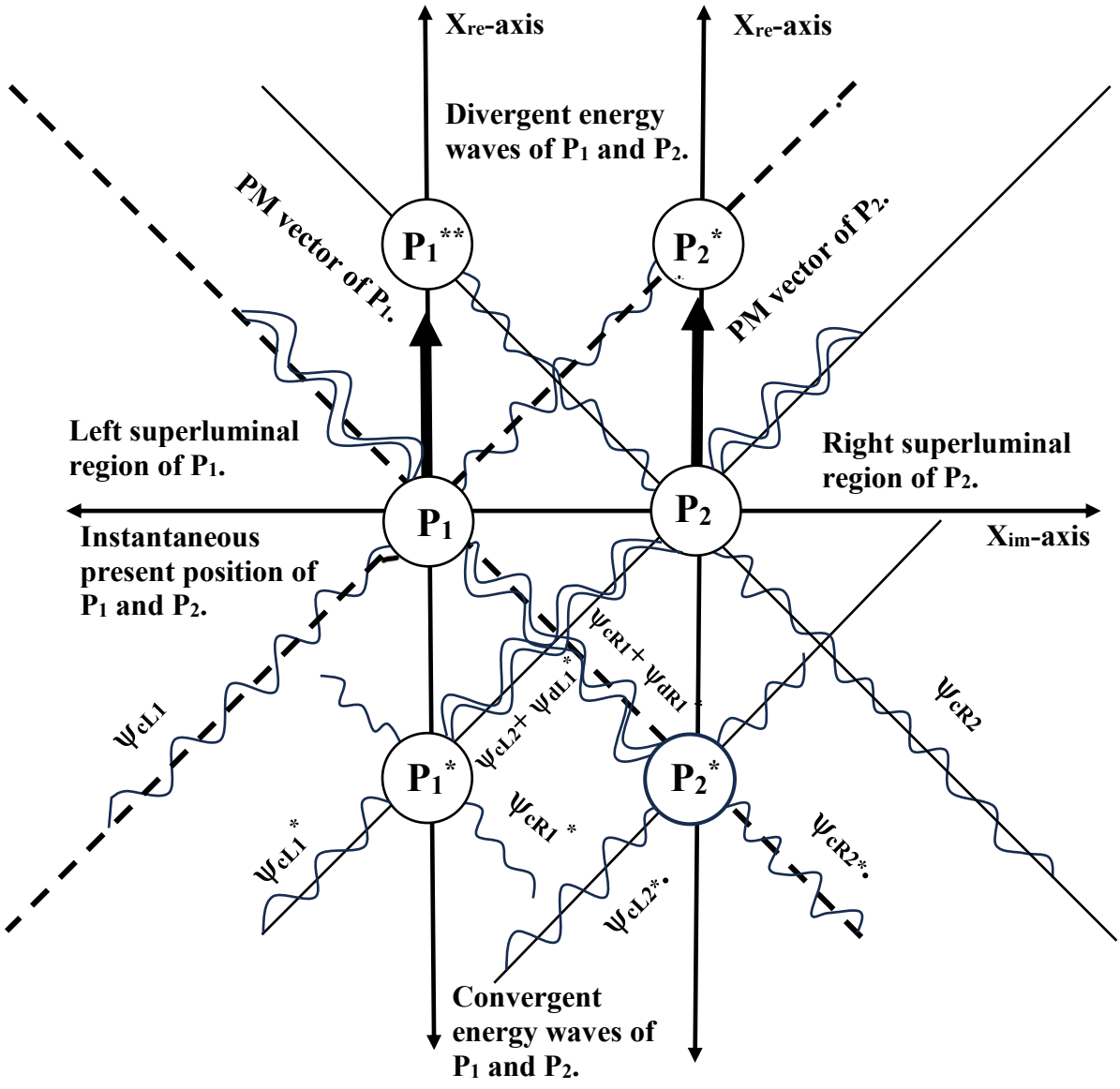


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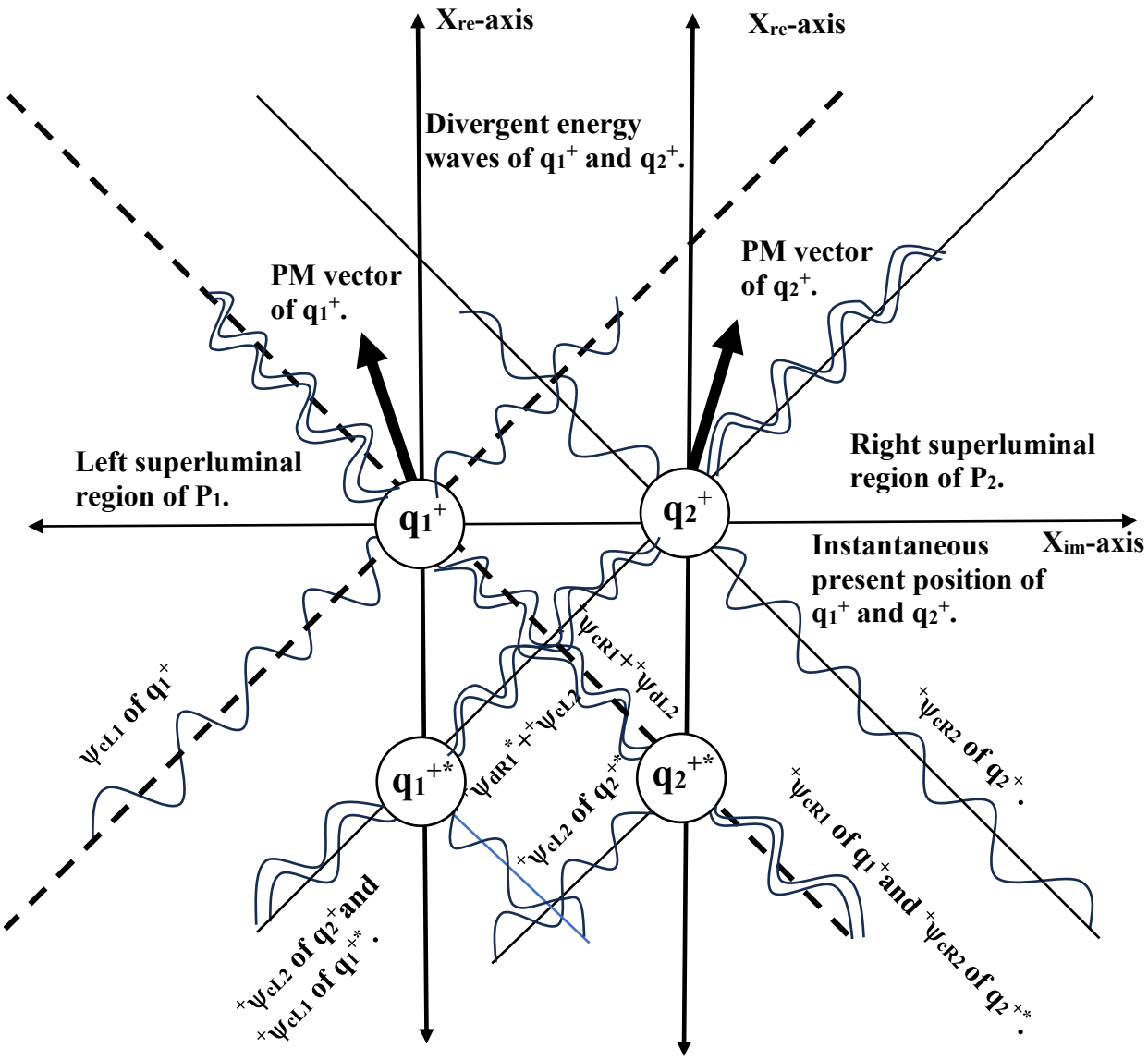


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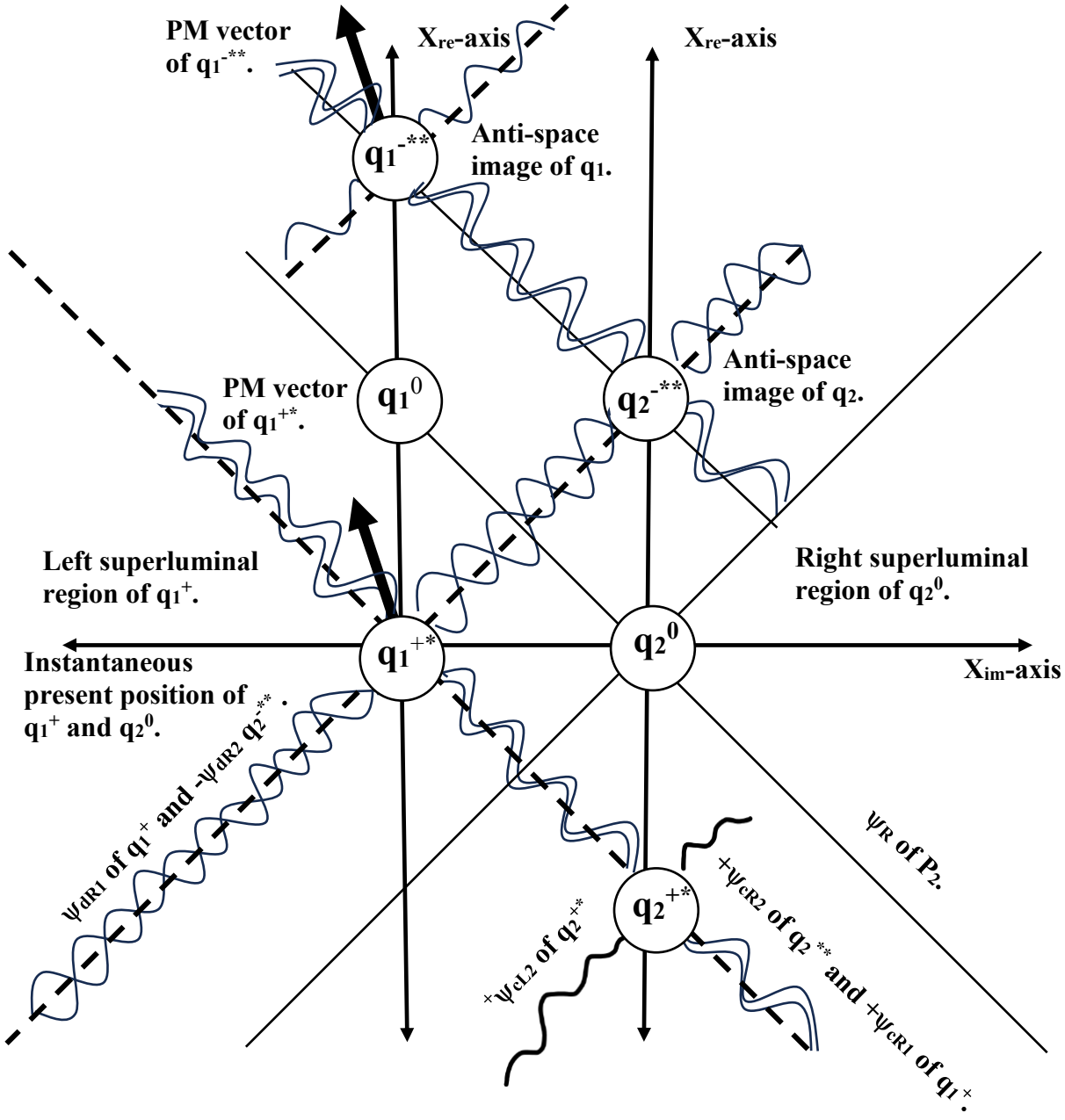


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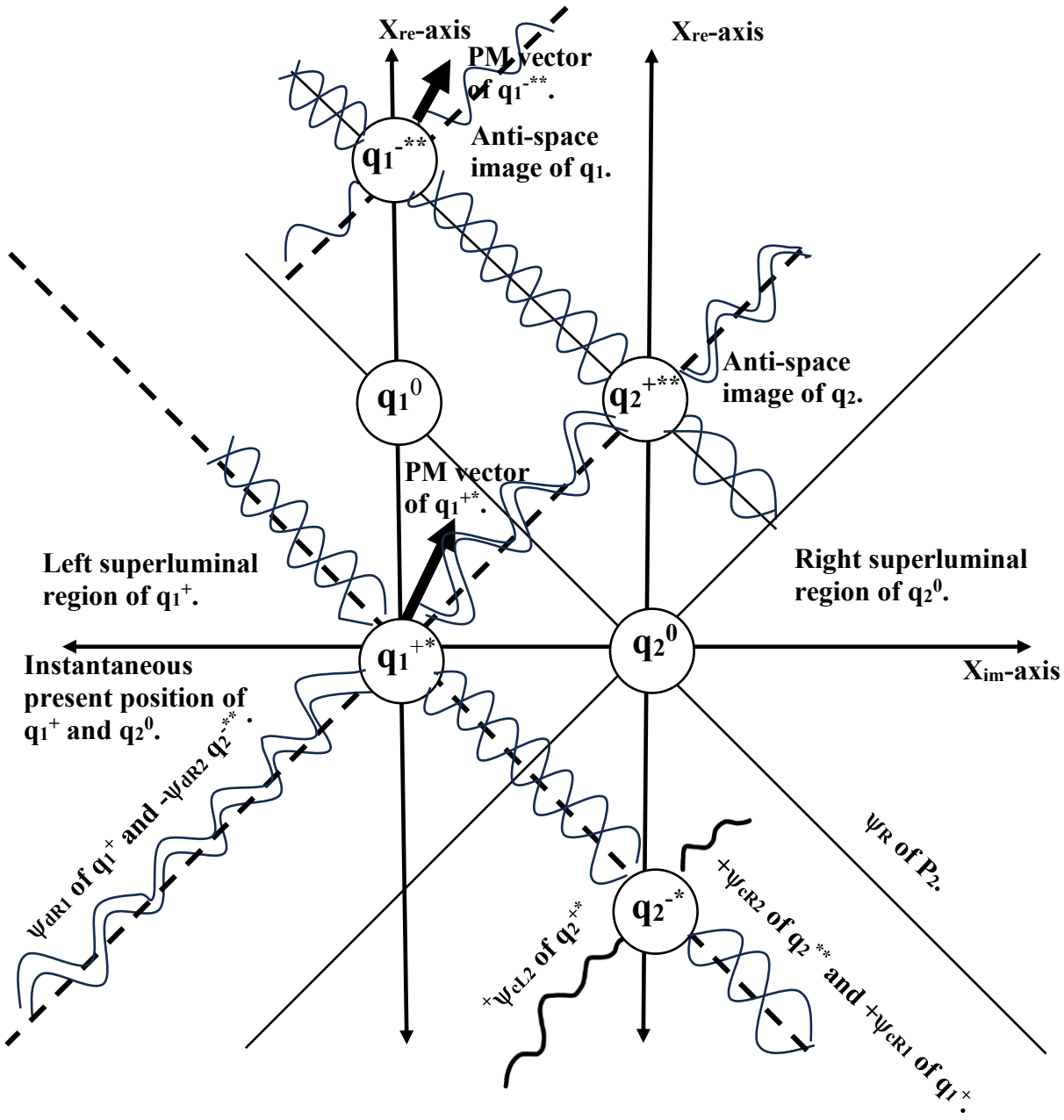


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